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Application of hierarchical thermo-mechanical modelling and development of a generalized heat-source trajectory for an aluminium WAAM component

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Abstract

Wire Arc Additive Manufacturing enables cost effective production of metallic components, however accurate prediction of the thermo-mechanical behaviour remains computationally demanding because of complex thermal interactions, repeated heating and cooling cycles and thousands of sequentially deposited beads. This work presents an application of a three-level hierarchical thermo-mechanical modelling framework for WAAM and its experimental validation through the fabrication of an aluminium circular nozzle-like component. The methodology consists of bead layer and wall level analyses, solving the thermo-mechanical phenomena at each stage for the subsequent pass to the next level. the present framework combines transient heat transfer and structural finite element analyses to predict temperature evolution, heat accumulation, residual stress development and geometric distortion throughout the manufacturing process. The complete particular wall level simulation consists of 27 deposited layers comprising 1,431 sequentially activated beads, requiring the solution of approximately 1.7 billion of equations and 1.6 TB output database. A generalized heat source trajectory formulation based on the deposition geometry is introduced to represent transient deposition behaviour. At the bead level, it can be represented by a constant position path, at the layer, by sinusoidal trajectory, and at the wall level, by an oscillated sinusoidal function with progressively decreased amplitude. Experimental validation using a robotic GMAW-CMT WAAM system has demonstrated excellent agreement between numerical and experimental results, with fabricated component closely resemble the desired geometry and temperature peaks deviation of only 1.05 %. The developed methodology provides a computational framework for automated WAAM process planning and optimization, establishing a basis for subsequent integration of real-time sensing and closed-loop process control.

Keywords Wire arc additive manufacturing · Finite element analysis · Thermo-mechanical analysis · Hierarchical modelling · Parallel computing

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Introduction

Additive manufacturing has emerged as a transformative manufacturing paradigm, enabling solutions to challenges that were previously difficult or impossible to address using conventional fabrication methods. Offering exceptional design freedom and fabrication flexibility, those qualities have driven its adoption in the aerospace, automotive, naval and biomedical industries[1]. Among numerous metal additive manufacturing (AM) technologies currently available, Laser Beam Fusion (LBF), Electron Beam Melting (EBM) and Wire Arc Additive Manufacturing (WAAM) are established as the predominant approaches for fabricating metallic components[2–4]. While LBF and EBM are capable of production of highly detailed parts with great dimensional accuracy, WAAM is characterized by high deposition rates, great cost-effectiveness and with one of the main advantages as the suitability for manufacturing large-scale structural components[5]. Because of this, WAAM is attracting considerable attention for applications that require high productivity and reduced material waste.

The historical development of WAAM can be traced back to the 1920s, when Baker has introduced and patented his process based on the layer by layer deposition of welding metal material[6]. Initially conceived primarily for decorative applications and subsequently largely overlooked, future technological advances gave it a possibility of the renaissance when in 1990s Cranfield University in collaboration with Rolls-Royce started the Shaped Metal Deposition program[4]. This has established the foundation for the modern metal AM technologies. From 2017 until to this day, WAAM has attracted a substantial research interest which can be seen with exponential publications growth and investment attraction to the technique.

Despite the growing interest to the topic, several fundamental challenges remain unsolved. Residual stresses accumulation, geometric distortion during deposition of melted material, micro-structural development and complete manual control of the manufacturing process among many, are continuing to limit printing reliability and manufacturing component performance, driving ongoing extensive research aimed at understanding underlying mechanism and developing effective mitigation strategies[7–9]. To address the issue of temperature accumulation a substantial amount of research has been conducted towards numerical simulation of the process[10–15]. Different numerical approaches have been employed to model the AM process, such as: Finite Element Method[16–19] and Computational Fluid Dynamics[20–22]. Although CFD provides a more detailed representation of the underlying physical phenomena, the astonishingly high computational cost, that sometimes reaches up to 15 days to simulate 6 s of the process, leaves FEA a predominant way of studying the thermal history and residual stresses generation of desired manufactured component[23]. For better understanding of inheritable problem of bead overlapping stochastic geometric overlapping bead model was proposed by S.G. Pulickan et. al.[24]. In addition to process-scale modelling, micro-structural analysis has also received considerable attention[25–28]. Investigation on how deposition strategies influence micro-structural development[29,30]. O P. Rosado et al. tried to combine metal deposition with sheet metal forming for creation of thin walled three dimensional components[31]. C. He et al. has studied the inter-layer stir friction processing for improvement of micro-structural evolution and mechanical properties of created builds[32], that has shown an effective elimination in pores and breakage of α-Al dendrites and Si-rich eutectic networks in specified Al-Si aluminium alloy. It is also worthy to specify that Al-Si is a great material that can be used in versatile fields of operation, although accompanied by ductility-strength paradox, which were investigated by B. Dang et al.[33]. Common strengthening strategies of Al-Si alloy are leading to inevitable ductility sacrifice, that can be somewhat prevented by micro-structural control of rapid solidification and past-solidification heat treatment. An interesting evolution of WAAM technique in addition of the secondary wire is proposed by U. Melkani et al.[34] that helps to increase already cost-effective deposition rate of the process by around 45%. An even more intriguing approach was proposed by L. Squires et al. that is exploring the fabrication of multi material structures though deposition of dissimilar metals using WAAM[35], drawing inspiration from naturally occurring structures. Singh et al. discussed possibilities of interrupted metal deposition technique[36], that provides better outcomes of the manufactured component. Biomedical innovations are going hand in hand with more implementation of AM into the field[37]. Orthopedics, dentistry, cardiology, and neurosurgery are among

few in which modern advances of 3D metal additive manufacturing are used[38]. Recent research has increasingly focused on incorporating machine learning[39] and artificial intelligence[40] into WAAM processes that might help to advance in-situ monitoring[41,42] and bring on the possibility of creation of the digital twin.

However, despite all the advanced research[43–45], automation remains one of the major challenges in WAAM and is yet to be resolved. Although significant steps forward have been made in process monitoring based on visual sensing[46] for parameters such as weld type identification, welding path generation and parameter planning, etc.[47], the results obtained are not providing the desired process quality. To decrease dependence on manual teaching and offline programming Y. Ma et al. introduced a framework for automated generation of welding path based on combined CAD and vision based data[48]. D. H. Ngoma et al. proposed robotic welding kit of six degrees of freedom for horizontal metal joints to diminish the complexity of traditional robotic systems and facilitate the manufacturing system installation[49]. R. Bolot showed the possibilities of modern slicing software and robot controlling[50]. Y. Yang et al. introduced a framework for fast construction of FE models for AM metal processes[51].

Despite substantial progress in WAAM automation and process monitoring, the integration of process-scale modelling, automated planning and feedback remains challenging. In particular, thermo-mechanical simulations of complete multilayer components can become computationally expensive because of the large number of sequential deposition events and repeated thermal and mechanical solution steps. This suggests that successful implementation of an automated WAAM framework may benefit from hierarchical rather than single-scale modelling strategies. In the present study, single-scale modelling refers to the direct simulation of the complete component, whereas the proposed multi-scale strategy divides the manufacturing process into three hierarchical levels: bead, layer and wall. Microscale model must capture the solidification behaviour, micro-structural evolution, melt pool dynamics and material transfer. The most suitable approach for the micro-scale is must be found in CFD simulations as considered by X. Chen et al.[52]. Mesoscale model, then is used to study layer formation, bead overlapping and temperature accumulation on adjacent bead. To more accurately capture the thermal gradients of the process the traveling steady pool approach developed by S.H. Ahn can be used[53]. On macroscale, the model can be reduced and simplified for considerable reduction in computational cost while retaining the ability to predict key manufacturing outcomes, primarily concerned with the evolution of the component geometry, heat accumulation throughout the build, and development of residual stresses and distortions. These approaches motivate a hierarchical study in which different levels of process representation can be combined according to the modelling objective. In the present study, this concept is implemented through three modelling levels - bead, layer and wall - using a consistent thermo-mechanical formulation and sequential transfer between levels.The conventional micro–meso–macro multiscale modelling of these physical phenomena is complementary to the proposed hierarchical strategy and is outside the scope of the present study.

To address the identified limitations, the present study extends the hierarchical thermo-mechanical modelling approach by introducing a generalized heat-source trajectory formulation that links the transient thermal response to the deposition geometry across multiple modelling levels. The main novelty lies in representing the evolution of the effective heat-source position directly from the reconstructed temperature field and expressing this evolution analytically at the bead, layer and wall scales. At the bead level, the trajectory is represented by a constant-position function; at the layer level, the periodic deposition geometry is represented by a sinusoidal function; and at the wall level, the trajectory is formulated as an oscillatory sinusoidal function with progressively varying amplitude associated with the evolving thermal response and changing deposition geometry. This formulation provides a unified mathematical description of heat-source evolution rather than treating the deposition trajectory independently at each modelling scale. In parallel, the hierarchical framework is implemented as an interaction-driven sequence in which the thermo-mechanical information obtained at the bead level is transferred to the layer level and subsequently to the complete wall level, enabling local process behaviour, bead interaction, heat accumulation, residual stress and distortion to be evaluated at their respective scales. The proposed methodology is experimentally demonstrated for an aluminium WAAM component using a robotic GMAW-CMT system, providing validation of both the predicted thermal response and the resulting component geometry. In this context, the proposed framework provides a computational basis for automated WAAM process planning

and optimization by sequentially evaluating bead-, layer-, and wall-level behaviour. The generalized heat-source trajectory further supports geometry-dependent deposition path generation, while real-time sensing and closed-loop adaptive control remain future developments.

Methodology

Numerical model construction

Numerical simulation of additive manufacturing processes for WAAM in particular can be used as a pre-processing stage. Requirements of which derived by its multilayer deposition characteristic. This enables systematic investigation of the process parameters that are governing component formation, providing insight into their influence of the quality and integrity of the final manufactured component. The primary objective is to reduce the occurrence of manufacturing defects and improve the reliability of the final structure. The WAAM process itself can be defined as direct energy deposition process in which a consumable wire a continuously melted by an electric arc and deposited later by later onto a substrate to fabricate metallic components. As successive layers are deposit, the evolving geometry and repeated thermal exposure give rise to complex heat accumulation and cyclic heating and cooling throughout the build. These transient thermal conditions govern melt-pool evolution, solidification behaviour inter-laying bonding and the resulting micro-structure, at the same time influencing the development of residual stresses and geometric distortion. Prior to fabrication numerical simulation helps to reduce experimental production cost, to short process development time and lower manufacturing cost while improving process reliability. Thermo-mechanical finite element analysis constitutes an efficient numerical framework for simulating the coupled thermal and mechanical phenomena governing the WAAM process. To accurately simulate the WAAM process, the governing equations describing the sequentially coupled thermal and mechanical behaviour must be established. The thermal phenomena is determined by partial differential form of heat conduction equation[8] Eq. 1:

$$\rho\, c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + q \tag{1}$$

where ρ is material density, c_p is specific heat capacity, T – temperature, k – thermal conductivity, q - heat source.

To accurately capture the transient thermal response, boundary conditions must be prescribed through convective and radiative heat exchange with the surrounding environment and externally applied heat flux and are defined as Eq. 2:

$$-k\nabla T \cdot \mathrm{n} = h\,(T - T_\infty) + \varepsilon\sigma\,(T^4 - T_\infty^4) + q_s \tag{2}$$

where T – is local surface temperature, T_∞ is the ambient temperature, h is a convection coefficient, ε is emissivity, σ is Stefan–Boltzmann constant and q_s is applied surface flux. The fourth power dependent radiative heat transfer term arises from Stefan-Boltzmann law, which states that the thermal radiation emitted by a surface is proportional to the fourth power of its absolute temperature. Because of high near molten pool temperatures, radiative heat losses become significant during cooling of recently deposited material, while moderate temperatures are dominated by convective effects.

To represent correctly the spatial distribution of heat input different heat source model is used. Among available approaches are:

Uniform heat distribution[54] Eq. 3,

$$q = \frac{\eta \cdot V \cdot I}{w \cdot h \cdot l} \tag{3}$$

where the total energy input is determined by welding parameters V and I being welding voltage and current respectively with parameter η that describes the loss of arc's electrical energy to the environment. The parameters w, h and l denote the width, height and length of the deposited bead. The uniform heat source model assumes that the applied heat is distributed uniformly throughout a prescribed area or volume, resulting in a constant heat flux within the heat source region;

Gaussian heat source[55] Eq. 4,

$$q\ (x,y) = \frac{3\eta VI}{\pi r^2} \cdot\ exp\left(-3\frac{x^2+y^2}{r^2}\right) \tag{4}$$

where r is radius of the applied heat flux, x and y are Cartesian spatial coordinates measured from the center of the heat source. This 2D model assumes that the heat spreading following the gaussian distribution, maintaining maximum flux at the center while decreasing exponentially with increasing radial distance;

and Goldak's heat model[56] Eq. 5.

$$q_{f/r}\ (x,y,z) = \frac{6\sqrt{3}f_{f/r}Q}{abc_{f/r}\pi\sqrt{\pi}} \cdot\ exp\left[-3\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c_{f/r}^2}\right)\right] \tag{5}$$

which consists of two overlapping ellipsoids being frontal and rear ones, where a is the length parameter of the ellipsoid, b is the width parameter and c is the penetration depth. f_f and f_r are the heat partition coefficients satisfying Eq. 6.

$$f_f + f_r = 2 \tag{6}$$

This model is able to accurately reproduce the asymmetric thermal energy distribution generated by welding arc, enabling more realistic representation of heat transfer, generating improved predictions of thermal history, residual stress evolution and geometric distortion.

Although the Goldak heat source is considered more physically realistic for representing arc heat input, the uniform heat source model was adopted in the present study to reduce the high computational cost. The hierarchical modelling framework prioritizes computational efficiency while retaining the ability to predict the global thermal response required for process-scale analysis. In addition, the uniform formulation preserves the total heat input based on the welding parameters and deposited bead dimensions, while avoiding the need for additional spatial calibration parameters required by more detailed heat-source models for sophisticated molten-pool simulations. Taking these considerations into account, the simplified uniform heat source provides an appropriate balance between model fidelity and computational performance.

The governing heat transfer equation is spatially discretized using the finite element method, Eq. 7. Time integration is performed using the generalized θ method.

$$\rho c_p \frac{T^{n+1}-T^n}{\Delta t} = \theta\left[\nabla\cdot\left(k\nabla T^{n+1}\right)+q^{n+1}\right]+(1-\theta)\left[\nabla\cdot\left(k\nabla T^{n}\right)+q^{n}\right] \tag{7}$$

In the present study, $\theta = 1$ is adopted, which corresponds to the implicit backward Euler, Eq. 8.

$$\rho c_p \frac{T^{n+1}-Tn}{\Delta t} = \nabla\cdot\left(k\nabla T^{n+1}\right)+q^{n+1} \tag{8}$$

This choice provides numerical stability for transient heat conduction problems since Crank-Nicolson is generally not preferred for welding because it lack L-stability and may produce oscillatory solutions in the presence of steep moving thermal gradients and strong nonlinearities.

Weak variational form of the heat governing equation is derived by application of the Galerking weighted residual method, Eq. 9:

$$\int_{\Omega} \delta T \rho \, c_{\rho} \frac{\partial T}{\partial t} d\Omega + \int_{\Omega} \nabla \delta T \cdot \, k \nabla T d\Omega = \int_{\Omega} \delta T q d\Omega + \int_{\Gamma_q} \delta T \bar{q} d\Gamma \tag{9}$$

The temperature field is approximated using FE shape functions, resulting in the semi-discrete matrix Eq. 10:

$$\mathrm{C}\frac{\partial \mathrm{T}}{\partial t} + \mathrm{KT} = \mathrm{Q} \tag{10}$$

where C is the global heat capacity matrix, K is the global thermal conductivity matrix, T is the vector of nodal temperatures and Q is applied internal heat associated with the applied welding heat source. Using implicit Euler scheme, receiving Eq. 11:

$$\mathrm{C}\frac{\mathrm{T}^{n+1} - \mathrm{T}^{n}}{\Delta t} + \mathrm{KT}^{n+1} = \mathrm{Q}^{n+1} \tag{11}$$

Rearranged to Eq. 12 is solved iteratively at each time increment to obtain temperature field throughout the deposition process:

$$\left(\frac{\mathrm{C}}{\Delta t} + \mathrm{K}\right)\mathrm{T}^{n+1} = \frac{\mathrm{C}}{\Delta t}\mathrm{T}^{n} + \mathrm{Q}^{n+1} \tag{12}$$

The primary output of the thermal analysis is the temperature field Eq. 13:

$$T = T(\boldsymbol{x}, t) \tag{13}$$

which serves as the thermal loading for the subsequent thermo-mechanical analyses used to predict residual stresses and distortions.

The mechanical behaviour of the WAAM is considered within continuum mechanics, where deposited material is considered as a continuous medium. Assuming quasi-static condition, the process is governed by Eq. 14:

$$\nabla \cdot \boldsymbol{\sigma} + \rho \boldsymbol{b} = 0 \tag{14}$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor, ρ is density and $\boldsymbol{b}$ is the body forces. The unknown variable in the mechanical problem is the displacement field Eq. 15:

$$\mathrm{u} = \mathrm{u}(\mathrm{x}, t) \tag{15}$$

Assuming small deformations, the kinematic relation between the displacement and strain field is expressed through Eq. 16:

$$\boldsymbol{\varepsilon} = \frac{1}{2}\left(\nabla \mathrm{u} + \nabla \mathrm{u}^{\mathrm{T}}\right) \tag{16}$$

The total strain is decomposed into elastic, plastic and thermal contributions[8], resulting in Eq. 17:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_{elastic} + \boldsymbol{\varepsilon}_{plastic} + \boldsymbol{\varepsilon}_{thermal} \tag{17}$$

The stress strain relationship is therefore described by thermo-elasto-plastic behaviour, Eq. 18:

$$\boldsymbol{\sigma} = \boldsymbol{D}(T) : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{plastic} - \boldsymbol{\varepsilon}_{thermal}) \tag{18}$$

where $\boldsymbol{D}$ is the elasticity matrix dependent on temperature field T. The mechanical response is described using a temperature-dependent von Mises elasto-plastic constitutive model with isotropic hardening and an associated flow rule. The yield stress is temperature-dependent and evolves with accumulated plastic strain.

Finite element discretizations is performed by expressing the equilibrium equation in its weak variational form using the Galerking method and integrating by parts, Eq. 19:

$$\int_{\Omega} \boldsymbol{\sigma} : \delta\boldsymbol{\varepsilon}\, d\Omega = \int_{\Omega} \delta \mathrm{u} \cdot \rho \mathrm{b}\, d\Omega + \int_{\Gamma_t} \delta \mathrm{u} \cdot \bar{\mathrm{t}}\, d\Gamma \tag{19}$$

Approximating the displacement field yields a mechanical response semi-discrete system of equilibrium equations, Eq. 20:

$$\boldsymbol{K}_M \boldsymbol{u} = \mathrm{F}_{external} - \mathrm{F}_{thermal} - \mathrm{F}_{plastic} \tag{20}$$

where $\boldsymbol{K}_M$ it's a global stiffness matrix, $\mathrm{F}_{external}$ - external force vector, $\mathrm{F}_{thermal}$ - thermal load vector and $\mathrm{F}_{plastic}$ - plastic contribution. Since deformations in WAAM processes are mostly internally driven $\mathrm{F}_{external}$ can be considered equal to 0.

The commercial finite element software Abaqus was used to perform the analysis. The finite element mesh is locally refined in regions directly involved in material deposition Fig. 6. The deposited beads and the adjacent regions are discretized using smaller elements. The mesh is progressively coarsened with increasing distance from the deposition region. In the analysis DC3D10 and C3D10 quadratic tetrahedral elements are used for substrate. For the thermal analysis DC3D20 elements are employed for bead region - twenty node quadratic heat transfer elements. For the mechanical analysis C3D20R elements with reduced integration are used for bead region. Although reduced elements using only 8 integration points instead of 27, it mitigates volumetric locking and numerical robustness during activation of deposited material using element birth and death technique, which was used for the sequential activation of the deposited material. A formal mesh-sensitivity study was not performed in the present work and is recognized as a limitation. For the transient thermal analysis, automatic adaptive time stepping was employed in Abaqus. Each bead deposition step had a total duration of 10 s, with an initial time increment of 0.1 s, a minimum increment of 10^{-5} s, and a maximum increment of 1 s. The maximum number of increments per step was set to 1000. The temporal discretization was selected consistently with the spatial discretization according to $v\Delta t \leq l_e$. For the refined deposition region, $l_e = 1.25$ mm and the heat-source displacement per increment was 1.17 mm.

Three-level hierarchical WAAM model definition

Previous studies have established thermo-mechanical and multiphysics approaches for analysing WAAM at different process scales[18–23,44,57,58]. However, these approaches generally focus on a specific modelling scale or on direct simulation of the complete component. A hierarchical workflow can instead use lower-level analyses to characterize deposition behaviour and screen process parameters before the computationally intensive wall-level simulation. Thermo-mechanical modelling, distortion prediction and melt-pool simulations further provide the numerical foundation for process optimization[18,20]. Unlike conventional hierarchical models which primarily distinguish levels according to geometric scale or mesh refinement, the present framework is interaction-driven, where each level is designed to evaluate process stability and transfer the resulting information to the subsequent level, enabling refinement of modelling assumption, deposition strategies and thermal management throughout the manufacturing process.

The present three-level hierarchical framework, illustrated in Fig. 1, consists of bead, layer and wall levels and is represented by three spatial domains:

$$\Omega_b \subset \Omega_l \subset \Omega_w \tag{21}$$

At the bead level the focus should be considered on molten-pool behaviour, geometry of the deposited bead, its penetration depth and thermal local deposition characteristics. Based on this information the possibility of going to the next level is concluded. At the layer level - bead overlapping, geometrical properties of layer formation, waviness of the surface, deposition trajectory, heat spreading to the adjacent beads and its total accumulation are evaluated. The final wall level integrates in itself prediction of heat accumulation, residual stress evolution,

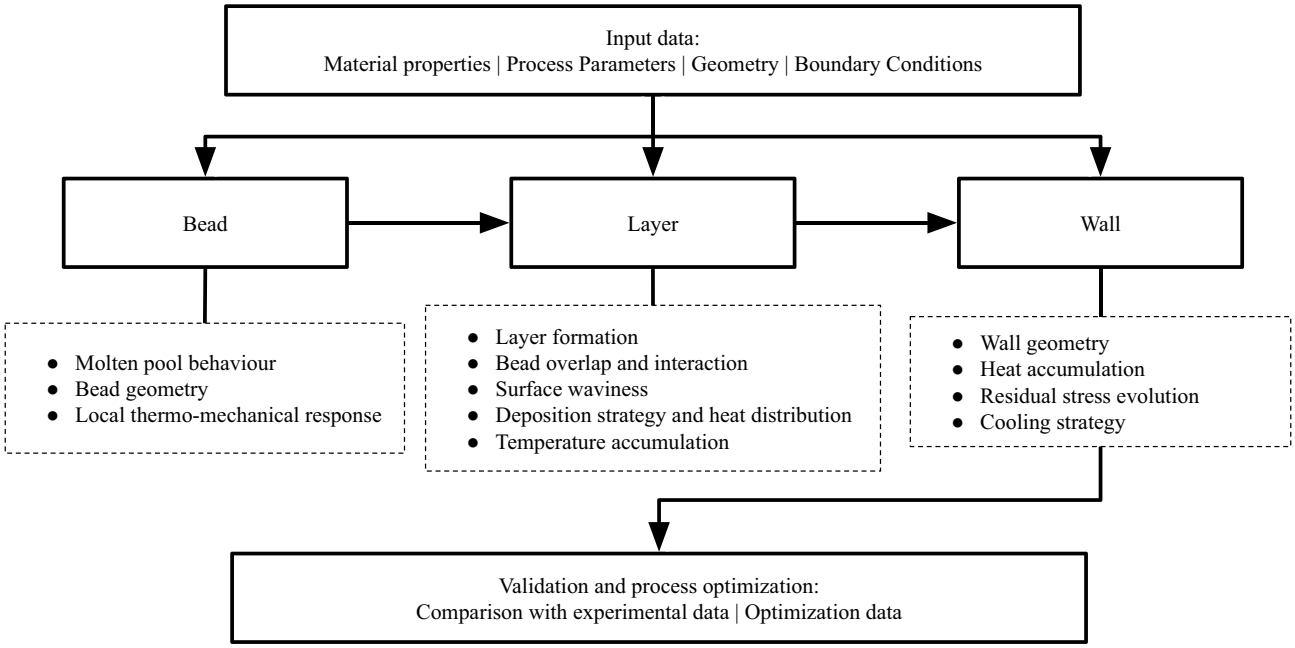


Fig. 1 Three-level hierarchical WAAM model framework.

geometric parameters and distortions. Information is transferred sequentially between the three levels, allowing each simulation to provide the input and the decision required for conduction of the next stage. The completed wall level analysis provides an automated computational basis to evaluate process feasibility, identify optimal process parameters and assess mitigation strategies before experimental manufacturing. For present work the physical formulation and numerical implementation remained consisted across all levels. Each hierarchical stage addressed a distinct modelling objective. The bead-level model characterized local heat input and bead geometry. The layer level evaluated heat accumulation and thermal evolution along path. The wall level captured cumulative thermal effects, residual stress and geometrical distortions. This hierarchical workflow reduced computational need compared to immediate full single-level simulations.

Geometrical and material parameters

The main objective of the research is to investigate the feasibility of manufacturing a hollow nozzle-like aluminium component with a tapered circular geometry using WAAM. The desired component must have an outer diameter of $D_b = 85$ mm at the base, $D_t = 65$ mm at the top over the final deposited layer and constant wall thickness $t_z = 5$ mm. The total projected height is $H = 67.5$ mm, Fig. 2. Such geometries are commonly found in rotationally symmetric engineering components used in aerospace, automotive, marine and energy applications.

A wide range of feedstock materials such as iron, aluminium, zinc, stainless steel and their alloys can be employed in the WAAM processes. Material selection determines not only the mechanical performance of the manufactured component but also the thermal behaviour during deposition, which directly influences microstructural evolution and the final properties of the desired specimen. Those properties include strength, hardness and the distribution of residual stresses. In the present work, aluminium alloy 6061 was selected as the feedstock material because of its balanced mechanical properties[59], excellent corrosion resistance and widespread use in key manufacturing technologies. For the subsequent simulations it is considered that both base-plate and the deposited bead are consisting of the same predefined alloy. The substrate and the deposited beads material properties are defined separately, Tables 1 and 2.

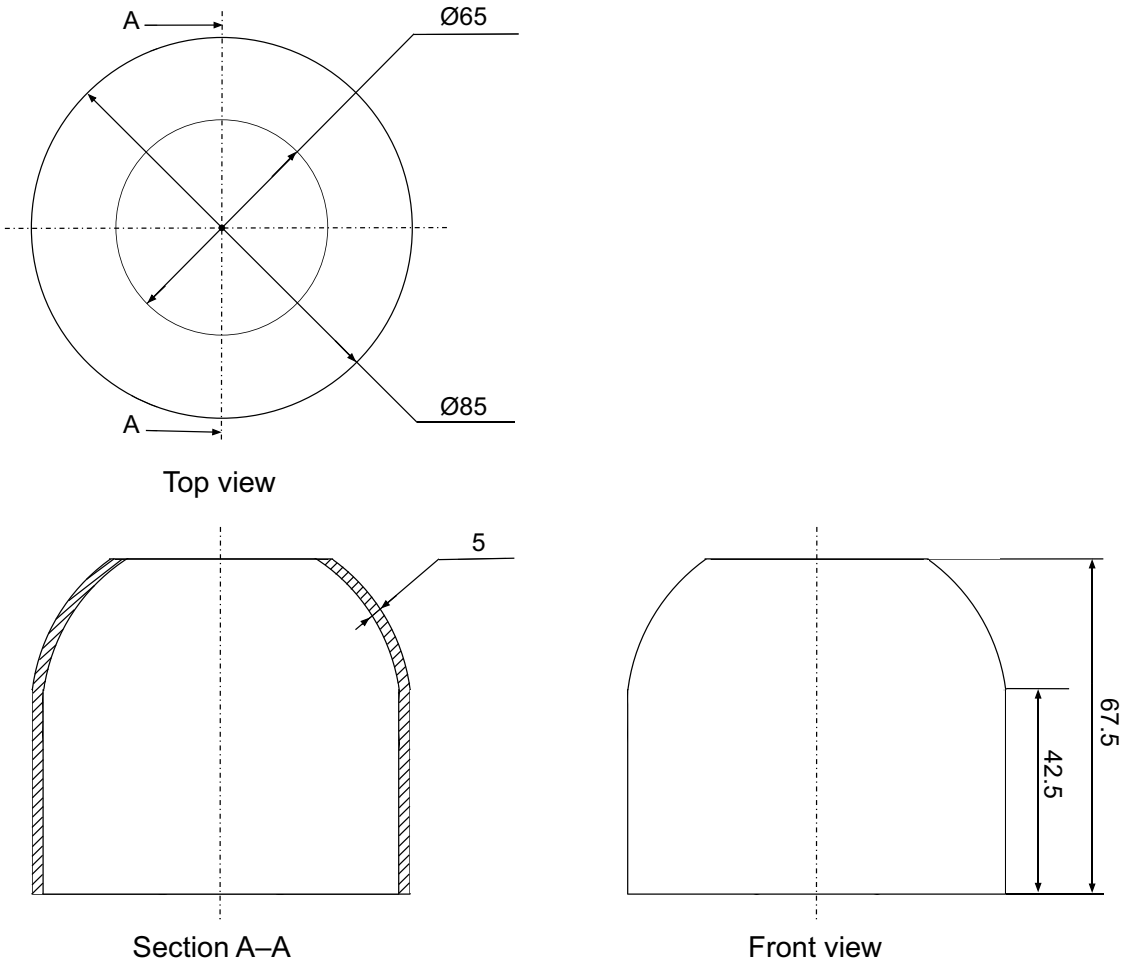


Fig. 2 Geometrical properties of the desired WAAM manufactured specimen showing the top, front and sectional (A-A) views. The principal dimensions are indicated in mm.

Table 1 Temperature dependent thermophysical and mechanical properties assigned to the deposited aluminium beads.

Thermal conductivity		Density	Elasticity			Thermal expansion coefficient		Heat capacity
T,°C	q, W/m C	ρ, kg/m³	T, C	σ, N/m²	ν	T, C	, C	c, J/kg C
300	164	2750	25	72.4e9	0.3	300	27.0e–6	900
500	194		500	63.5e9		500	27.4e–6	
700	210		700	56.1e9		700	27.7e–6	
770	220		770	50.0e9		770	28.3e–6	

Table 2 Thermophysical and mechanical properties assigned to the aluminium substrate.

Thermal conductivity	Density	Elasticity		Thermal expansion coefficient
q, W/m C	ρ, kg/m³	σ, N/m²	ν	, C
204	2750	69e9	0.33	2.3e–5

This is done because only the deposited material requires temperature dependent properties. During the printing process, the deposited bead experiences temperatures of approaching the alloy melting point (approximately 660°C) where pronounced nonlinear variations in its thermal and mechanical properties occur. In contrast, the base-plate remains within a substantially lower temperature range and serves only as a supporting substrate rather than as part of the final manufactured component. Therefore, the base-plate is presented using simplified temperature-dependent properties, unlike the deposited material with fully temperature-dependent thermal and mechanical properties.

Geometrical model of the first level of the three-level hierarchical WAAM model

To initiate the first level of the three-level hierarchical WAAM model, the geometrical parameters of the substrate, also called "base", on which deposition process if performed were defined. The substrate and bead dimensions were determined based on the target geometry of the desired manufactured component. A bead width of $w = 5$ mm and a height of $h = 2.5$ mm were selected. The corresponding welding parameters of $V = 16.1 \pm 0.5$ V and $I = 201 \pm 5$ A, with thermal efficiency coefficient set to 0.75. The substrate dimensions were defined as $L_{base} = 250$ mm in length, $W_{base} = 115$ mm in width and $V_{base} = 10$ mm in thickness, Fig. 3. To prevent unrealistic heat accumulation resulting from the finite substrate dimensions, a heat sink boundary condition representing heat dissipation to the surrounding environment was imposed on both lateral faces (110 mm) of the substrate by maintaining a constant ambient temperature of $T_{\infty} = 21.1$°C throughout the simulation. The mushy zone and latent heat were not explicitly modelled; instead, the thermal response was represented using temperature-dependent material properties. This simplification was adopted to maintain computational efficiency.

Geometrical model of the second level of the three-level hierarchical WAAM model

Following the bead-level analysis, the second level of the hierarchical WAAM model representing an individual level is considered. At this level, the deposition strategy consists of a circular trajectory composed of 53 sequentially deposited beads forming a closed contour with an outer diameter of $D_b = 85$ mm, Fig. 4. The trajectory was generated by discretizing the circular path into individual bead deposition events while preserving the prescribed bead geometry and process parameters defined at the first level. This enables the investigation of thermal accumulation, heat transfer between adjacent beads and the resulting thermo-mechanical response during the deposition of a complete layer.

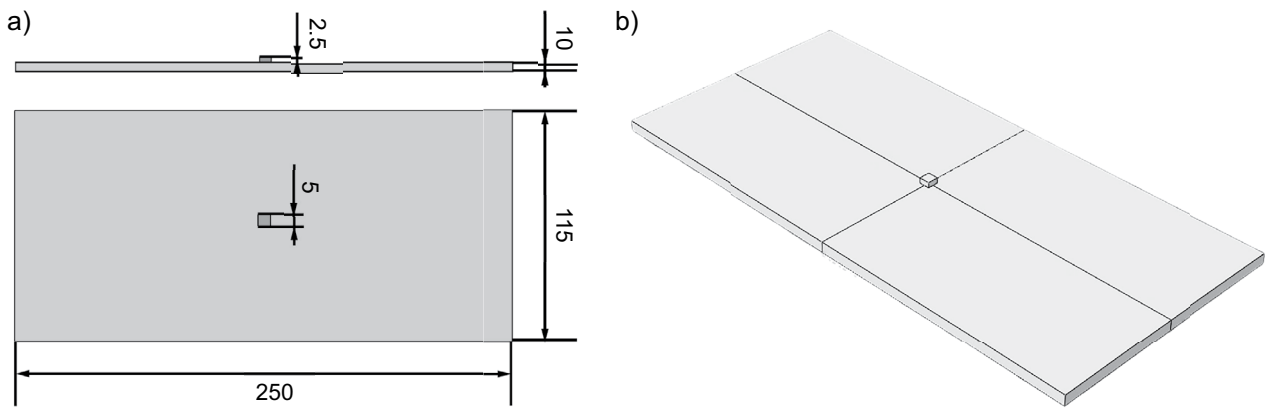


Fig. 3 Geometrical definition of the first level of finite element model employed in the hierarchical WAAM framework for thermo-mechanical analysis. (**a**) Substrate and deposited bead dimensions. Front and top view. (**b**) Isometric representation of the computational geometry for the bead level.

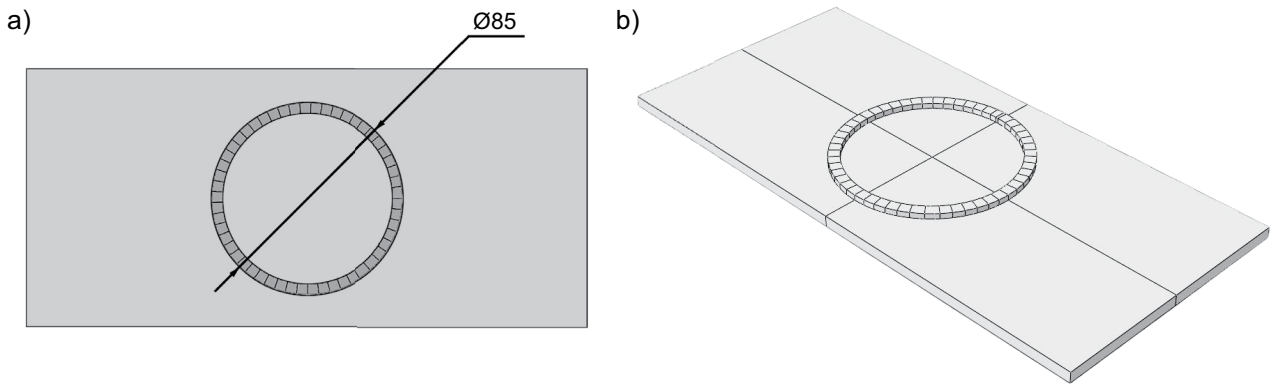


Fig. 4 Geometrical definition of the second level of finite element model employed in the hierarchical WAAM framework for thermo-mechanical analysis. (**a**) Circular deposition trajectory consistent of 53 sequential beads with outer diameter of $D_b = 85$ mm. Top view. (**b**) Isometric representation of the computational geometry for the layer level.

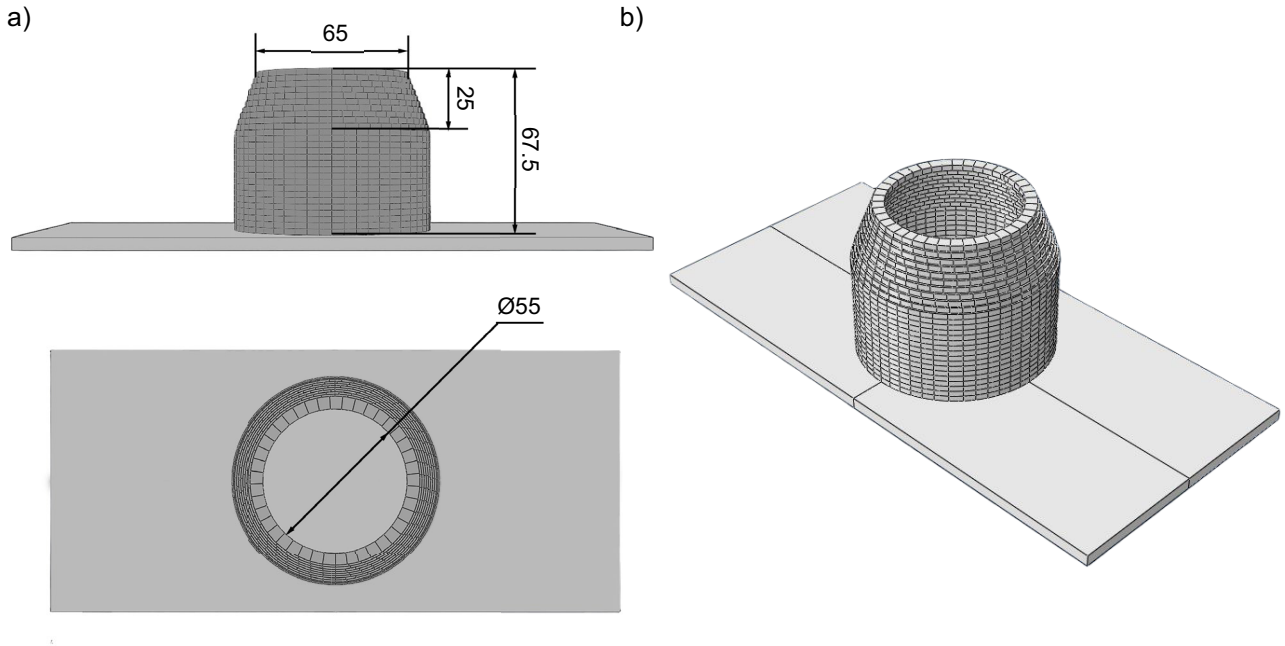


Fig. 5 Geometrical definition of the third level of finite element model employed in the hierarchical WAAM framework for thermo-mechanical analysis. (**a**) Geometrical dimensions of the final cylindrical WAAM component, consisting of 27 deposited layers and 1,431 sequentially activated beads. The inclined section is approximated by discrete $R_{offset} = 1$ mm radial steps, resulting in stepped profile. Front and Top view. (**b**) Isometric representation of the computational geometry for the wall level.

Geometrical model of the third level of the three-level hierarchical WAAM model

Building upon the results obtained from the bead and layer level simulations, the third level of the hierarchical WAAM framework was established to represent the complete cylindrical nozzle like component. The computational geometry was defined according to the dimensions of the manufactured part, consisting of an outer top diameter of $D_t = 65$ mm, an inner top diameter $D_{t_i} = 55$ mm, a total height $H = 67.5$ mm and an upper inclined section with a height of $H_{inclined} = 25$ mm, Fig. 5. The wall level model consists of 27 deposited layers comprising a total of 1,431 sequentially activated beads, accurately reproducing the complete manufacturing process and the

final component geometry. The inclined section was represented using a layer step approximation. Beginning with 17th layer, the outer diameter $D_{b_{layer}}$ of the component was reduced by $L_{step} = 2$ mm every layer, producing successive $R_{offset} = 1$ mm radial offset to achieve desired manufactured component geometrical dimensions. The computational implementation of the wall level model required 2,734 transient thermal analysis steps followed by 4,099 mechanical analysis steps, accounting for the sequential activation of each deposited bead and the corresponding thermo-mechanical response. The fully activated finite element model comprised 23,435 active elements, 85,345 active nodes and 252,612 equations. In cumulative terms, this corresponds to more than 1.7 billion equation solves throughout the complete simulation. Due to the large number of deposited beads and analysis steps, the wall level model represents the most computationally demanding stage of the present hierarchical WAAM framework. The simulation enables the prediction of the global temperature evolution, thermal gradients, residual stresses, and deformation. throughout the complete manufactured component.

Data extraction methodology

To enable quantitative evaluation of thermo-mechanical response, a consistent data extraction strategy was created and adopted throughout the three hierarchical WAAM simulation levels. Monitoring points for the substrate - "B" and for deposited welding beads - "WB" were defined separately for each level of the framework, Fig 6. The substrate monitoring points "B" were positioned closer to the edges of the base to evaluate heat dissipation

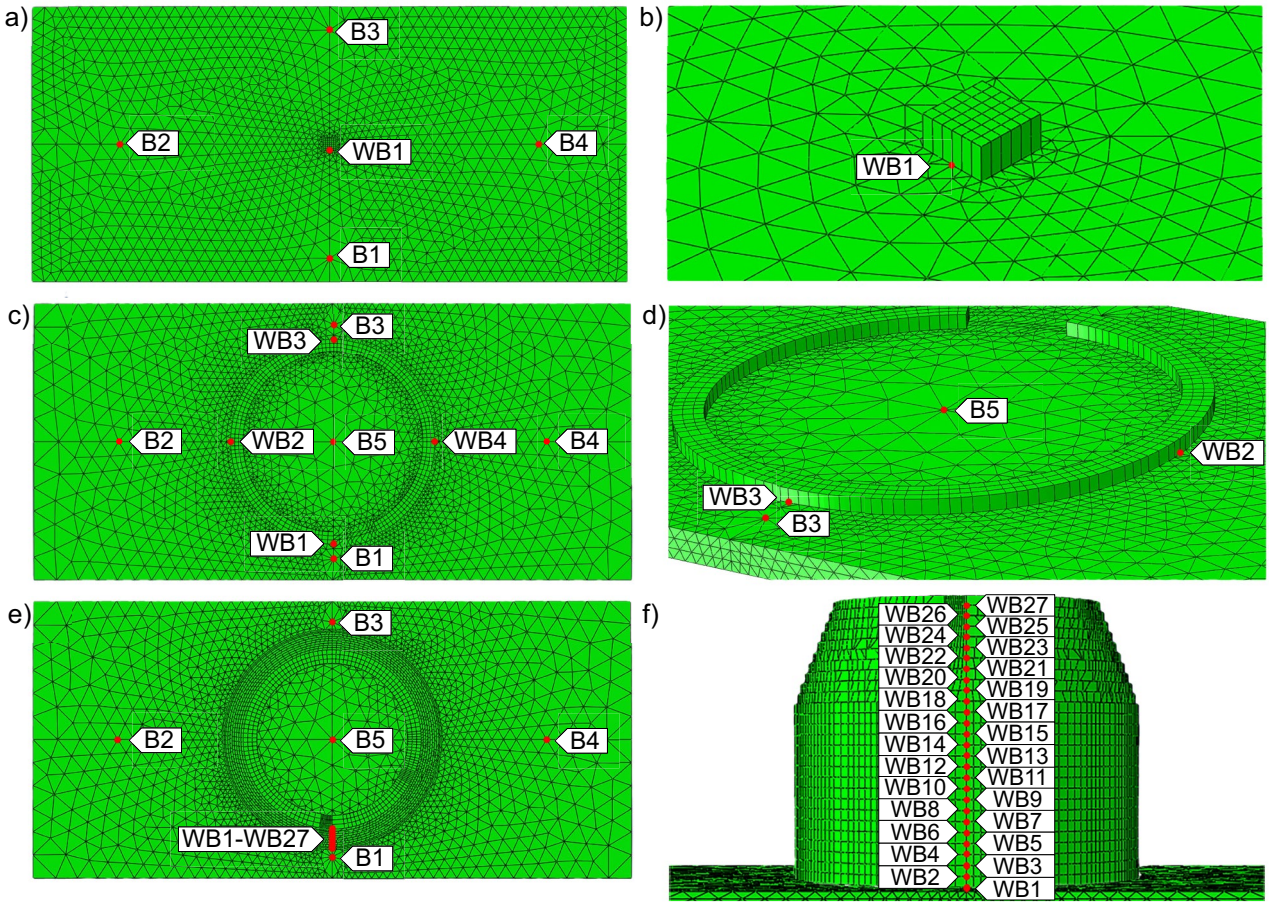


Fig. 6 Mesh discretization and temperature data extraction points adopted for the three hierarchical WAAM framework simulation levels. (**a**, **b**) Bead-level. (**c**, **d**) Layer-level. (**e**, **f**) Wall-level. Points named "B" denote base monitoring points, "WB" - welding beads monitoring points.

during the deposition process. Since, the first level model consists of a single deposited welding bead situated at the center of the substrate, no central substrate gathering point (B5) was defined at this level, whereas B5 was included in the layer and wall levels. The welding beads monitoring points were positioned at the bead-substrate interface. At the layer level, four welding bead monitoring points (WB1-WB4) were distributed uniformly along the deposited curvature to evaluate temperature distribution around the deposited layer. At the wall level, welding beads monitoring point was assigned to the starting location of each deposited layer (WB1-WB27), enabling the evaluation of global heat accumulation and thermal evolution throughout the manufacturing process. In addition to monitoring points, dedicated extraction path were defined according to the hierarchical simulation level and the physical quantity being investigated. These paths were selected to facilitate the post-processing of thermal and mechanical field variables and are introduced togheter with the corresponding results in the following sections.

Experimental robotic WAAM setup

Experimental validation was preformed using a robotic WAAM system based on a YASKAWA AR1440 industrial robot. The component was manufactured using the same deposition strategy, geometry and process parameters adopted in the numerical simulation, Table 3, enabling direct comparison between the experimental and numerical results.

The thermal response during deposition was monitored using a Bosch GIS 1000°C Professional infrared camera, with operational diapason of -40 to 1000°C and an infrared temperature sensor with diapason of - 20 to 1200°C. The camera was positioned at a fixed distance and viewing angle relative to the deposited wall to continuously monitor the same surface region during the manufacturing process. Temperature measurements were acquired at a predefined location corresponding to the WB1. The emissivity was selected according to aluminium at elevated temperatures and maintained constant during the experiments. The measured temperature histories were compared with the corresponding numerical predictions obtained at the equivalent monitoring location. The experimental procedure provided the thermal data required for validation of the hierarchical thermo-mechanical model.

Results and discussion

First level numerical results

The first level of the hierarchical WAAM model characterizes thermal response associated with a single deposited bead. The simulated thermal field Fig. 7 exhibits a pronounced temperature gradient centered right underneath the heat source, with a peak temperature of 660°C. The temperature decreases rapidly with increasing distance from the heat source, producing concentric isotherms characteristic of conductive heat dissipation. The thermal field is presented as both a longitudinal cross-sectional view and a top view of the deposited bead Fig. 7a. The cross-sectional view highlights the localization of the heat source within the deposited material, revealing the high temperature region where temperature exceed the melting threshold, defining the molten pool geometry, providing an estimate of the bead penetration depth, which represents effective fusion between deposited material and the

Table 3 Material deposition parameters.

Type of welding	GMAW CMT
Welding condition	PA (1G)
Electrode extension	15±2 mm
Welding speed	70 cm/min
Wire feed speed	10.0±1 m/min
Welding current	201±5 A
Voltage	16.1±0.5 V
Gas	15±1 l/min

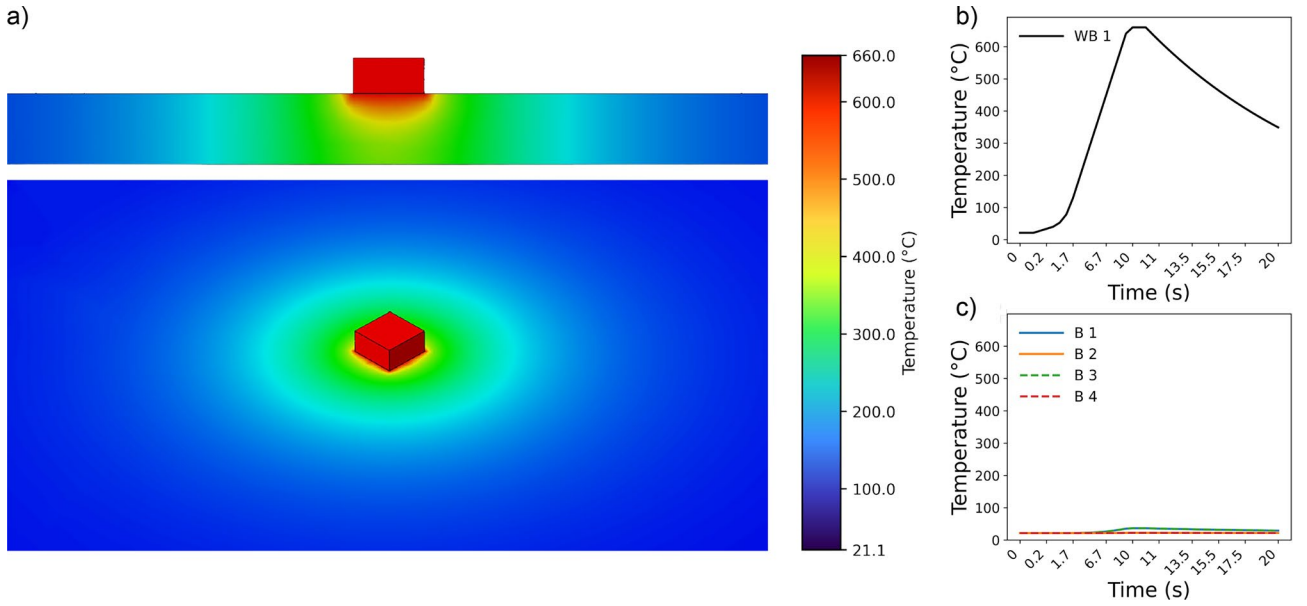


Fig. 7 Thermal response of the first level hierarchical WAAM model. (**a**) Predicted temperature distribution in the deposited bead shown as a longitudinal cross-sectional view, identifying molten region beneath the bead, and a top view, illustrating heat propagation into the substrate and the resulting thermal gradients. (**b**) Temperature history recorded at the welding bead monitoring point WB1, showing deposition and cooling process. (**c**) Temperature history recorded at the substrate monitoring points B1-B4, presenting relatively steady temperature levels.

substrate. The top view of Fig. 7a demonstrates the radial propagation of heat into the substrate through concentric temperature contours. Fig. 7b presents the temperature history gathered at WB1 monitoring point, 6b, showing a rapid increase in temperature during deposition followed by gradual cooling after deposition process has ended. In contrast, Fig. 7c, is representing base monitoring points B1-B4, Fig. 6a that are receiving only limited amount of temperature increase throughout the process. Comparatively low temperatures at these locations are due to relatively long distance to the heat source and quick dissipation of heat through convection and radiation. However it is possible to see that those beads that are situated closer to the deposition location are experiencing slightly higher thermal values, although negligible to influence structural response.

To obtain a more comprehensive understanding of the transient thermal behaviour during bead deposition, the temperature field was extracted along the predefined path Fig. 8a. Extracted data corresponding to all frames of all steps generated during analysis were combined and superimposed, producing the transient temperature profiles presented in Fig. 8b. This 2D profiles help to interpret evolution of the thermal field along the substrate during the deposition process. To better visualize the spacial-temporal history, the extracted profiles were reconstructed as a three dimensional surface, Fig. 8d, where the spatial and temporal development of the heat distribution can be visualized simultaneously. The resulting surface clearly shows the propagation of the thermal peak together with the associated temperature gradients generated. A top view projection of this surface, Fig. 8c, gives more possibilities to understand the evolution of the maximum temperature over time. The maximum temperature trajectory was extracted to define the heat source trajectory, Fig. 8e. For the first level model or the bead-level model, the deposited bead is located at the center of the base. According to this, the position of the maximum temperature remains constant in the transverse direction during the deposition process. Redefining the trajectory for the first level mathematically, we are getting:

$$x_b(t) = x_0 \tag{22}$$

where x_b is the temperature peak along path over first level of three-level hierarchical model over time t, x_0 is the position of the deposited bead. Since the bead is deposited at the center of the substrate:

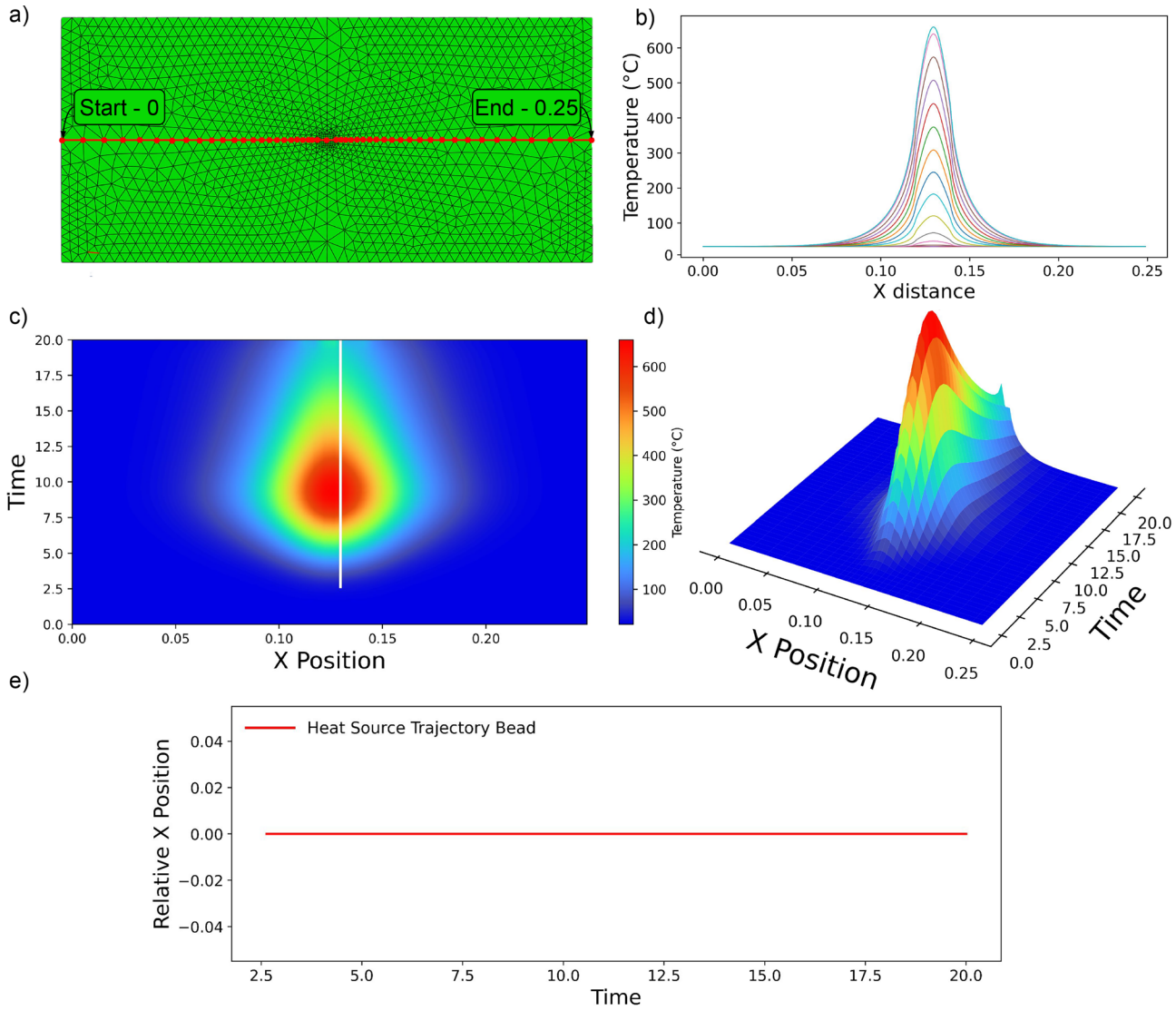


Fig. 8 Extraction and reconstruction of the transient temperature field for the first level hierarchical WAAM model. (**a**) Predefined extraction path along the substrate used to obtain data. (**b**) Temporal evolution of the thermal field along the selected path across all steps and frames. (**c**) Top view projection of the reconstructed spatio-temporal temperature surface with highlighted maximum temperature path. (**d**) Three-dimensional reconstruction of the transient temperature field, showing the propagation of the thermal peak and the associated temperature gradients during deposition. (**e**) Heat source trajectory obtained by tracking the position of the maximum temperature over time at the bead-level model.

$$x_b(t) = x_0 = 0 \tag{23}$$

These equations provide a mathematical description of the first level heat source trajectory within proposed three-level hierarchical WAAM model. Establishing this analytical representation enables a direct comparison with the corresponding trajectories obtained at the layer and wall levels, where increasingly complex deposition paths are expected. Such comparison provide the bases for evaluating whether a generalized trajectory formulation can be established for different deposition geometries.

The structural analysis at the first level of the three-level hierarchical WAAM model established the fundamental thermo-mechanical response associated with the deposition of a single welding bead. Data over path defined at Fig. 8a has been used for extraction of data. The transient thermal cycle generates a highly localized residual stress field that remains constrained to the bead-substrate interface, Fig. 9. The equivalent stress distribution,

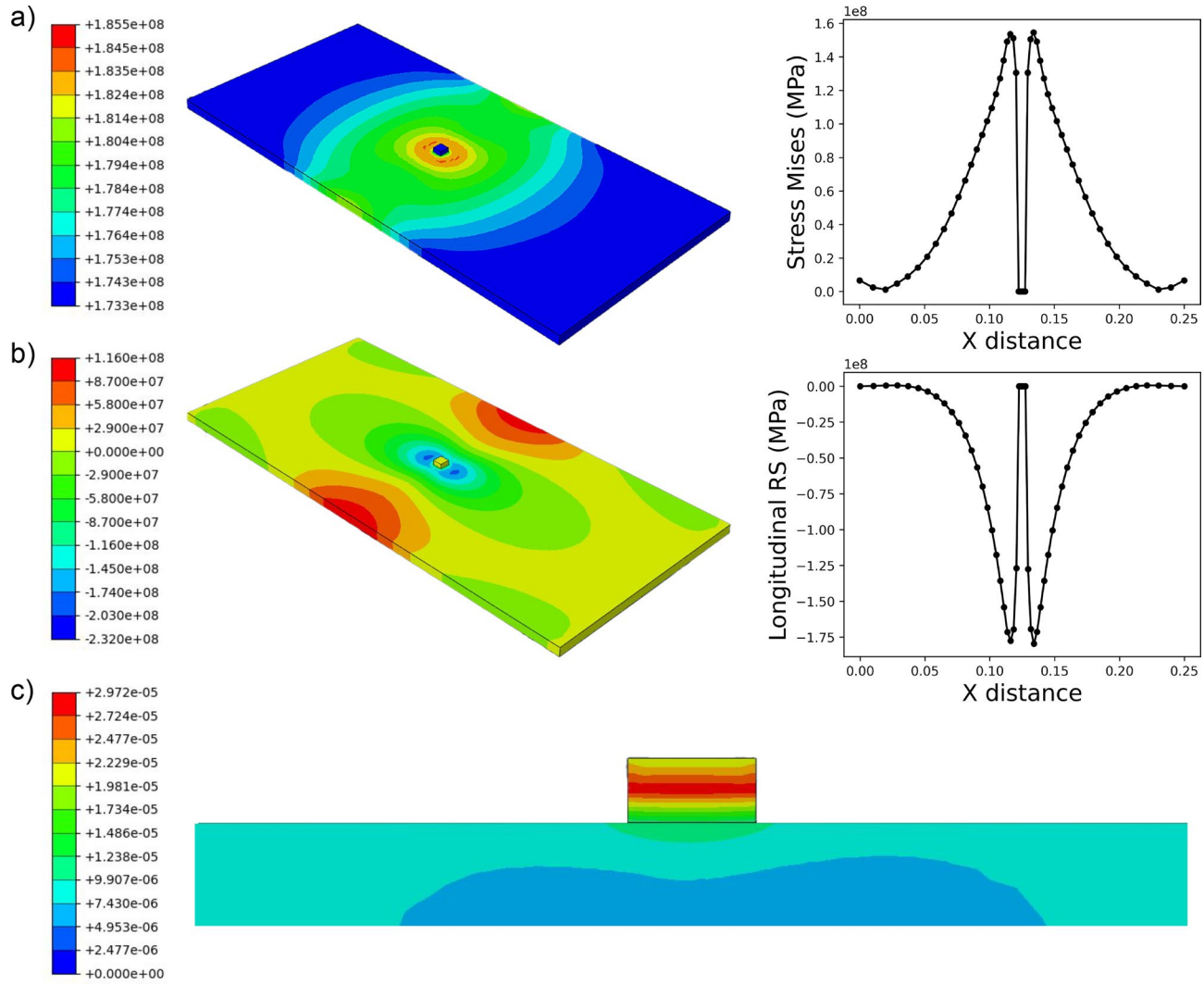


Fig. 9 Thermo-mechanical response of the first level hierarchical WAAM model. (**a**) Equivalent von Mises stress distribution together with the corresponding stress profile extracted along the predefined substrate path. (**b**) Longitudinal residual stress distribution and the corresponding path profile. (**c**) Displacement magnitude of the bead-level model demonstrating localized response.

Fig 9a reveals that the maximum stresses coincide with the region of the highest thermal gradients, confirming that residual stress formation is governed by localized thermal expansion and restrained contraction during cooling. The corresponding stress profile demonstrates a rapid reduction in stress magnitude with increasing distance from the deposition region, indicating that the mechanical influence of an individual bead is spatially limited. The longitudinal residual stress distribution, Fig. 9b is supporting that the substrate provides sufficient mechanical restraint to generate compressive stresses beneath the deposited bead while limiting the extent of the affected region. The symmetry of both the contour distribution and the extracted stress profile reflects the symmetric thermal loading associated with a single centered bead. The displacement field, Fig. 9c confirms that deformation remains localized within the deposited material and the adjacent heat affected zone, while surrounding regions experience only minor elastic deformation. The confinement of the thermo-mechanical response to the localized influence of the deposited bead demonstrates that the governing mechanisms can be resolved at the first level of the hierarchical WAAM model, enabling its systematic extension to the higher geometric levels.

Second level numerical results

The second level hierarchical model provides insight into the transient thermal evolution during the deposition of a complete circular layer, Fig. 10. Representative temperature distribution Fig. 10a-c are demonstrating the progressive expansion of the thermal energy as successive welding beads are being deposited, in contrast to the first level model, where the thermal response is governed by a single localized heat source. At Step 12, Fig. 10a, the initial deposition stage, temperature remains highly localized around the moving heat source and heat is conducted predominantly into the substrate and resembles more bead-level model. By Step 55, Fig. 10b, approximately at the half completed process of the circular deposition, thermal fields generated by neighboring beads begin to overlap, reheating previously deposited material and reducing local cooling rates. This causes the high-temperature region to expand, abolishing its purely localized behaviour. Near final stage, at Step 100, Fig. 10c deposition path reaching the starting point, succeeding in remelting of initially deposited material. Therefore, the temperature distribution is significantly influenced by the residual thermal energy retained within the previously deposited material, developing spatially heterogeneous thermal gradients. Temperature histories, Fig. 10d at each monitoring welding beads points WB1-WB4, Fig. 6c, exhibit rapid temperature raise during the time of near deposition event and rapid cooling interrupted by continuous deposition of the next beads, showing that thermo-mechanical behaviour now governed not only by the instant heat input buy also by the cumulative thermal history. The substrate monitoring points B1-B5 Fig. 6c exhibit a fundamentally different thermal response, Fig. 10e. Greater thermal mass and prescribe ambient temperature boundary conditions of 21.1°C along lateral sides are limiting excessive heat accumulation. The only difference is shown by central point B5, Fig. 6d, that gradually

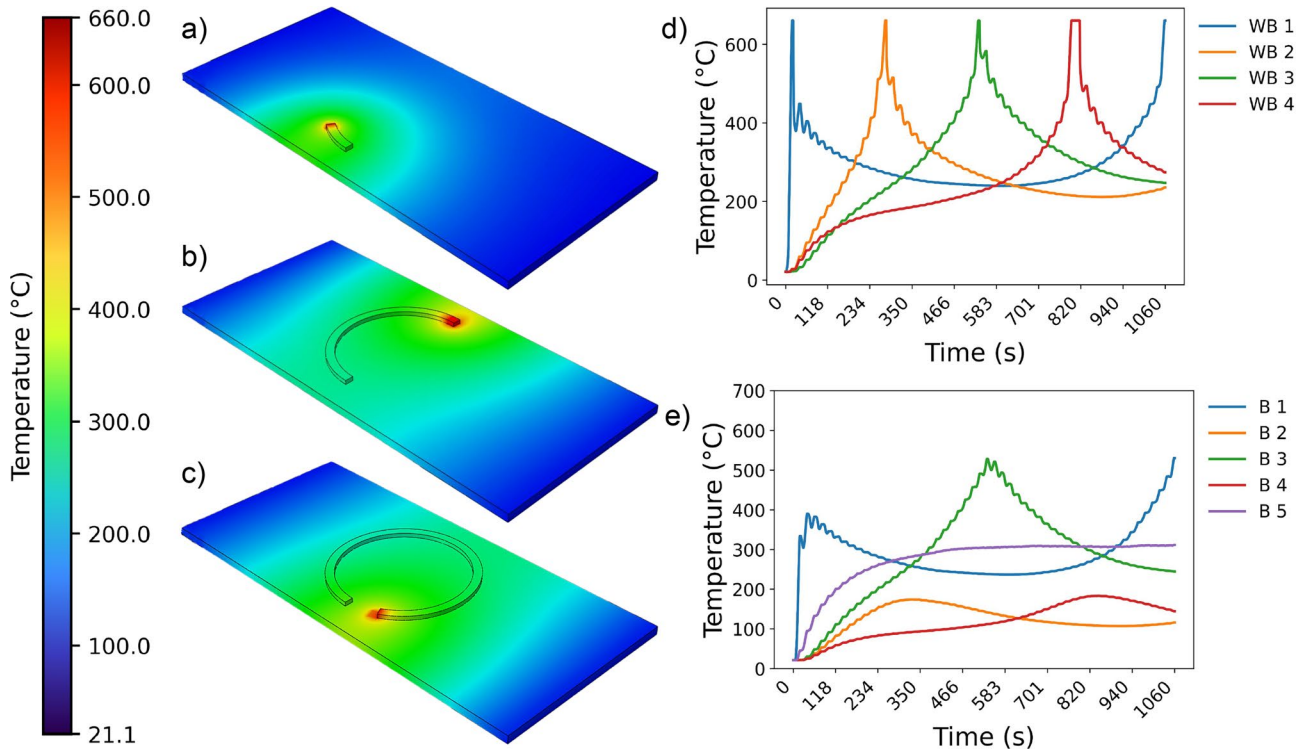


Fig. 10 Thermal response of the second level hierarchical WAAM model. Predicted temperature distribution of layer deposition process at Step 12 (**a**), Step 55 (**b**) and Step 100 (**c**) showing progressive evolution of the thermal field during sequential bead deposition. (**d**) Temperature histories recorded at the welding beads monitoring points WB1-WB4 localized in different sections of the layer geometry, presenting heat spread into adjacent beads and gradual accumulation. (**e**) Temperature histories recorded at the substrate monitoring points B1-B5, demonstrating gradual development of heat accumulation and thermal dissipation.

accumulating temperature continuously because of its localization, but not reaching melting points. The substrate temperature develops more smoothly and remains lower than that of the deposited layer effectively dissipating heat.

To investigate the influence of deposition geometry on heat source evolution trajectory, the transient temperature field of the second level model was reconstructed using the same methodology employed for the first level analysis. Temperature data were extracted along the predefined substrate path shown in Fig. 11a. All extracted temperature profiles were plotted, Fig. 11b. Unlike the first level model, where a single temperature peak is observed, the second model exhibit two distinct temperature peaks corresponding to two intersections between the extraction path and the circular deposition trajectory. These maximums are evolving continuously throughout the deposition process, reflecting the movement of the heat source along the circular path. The reconstructed three

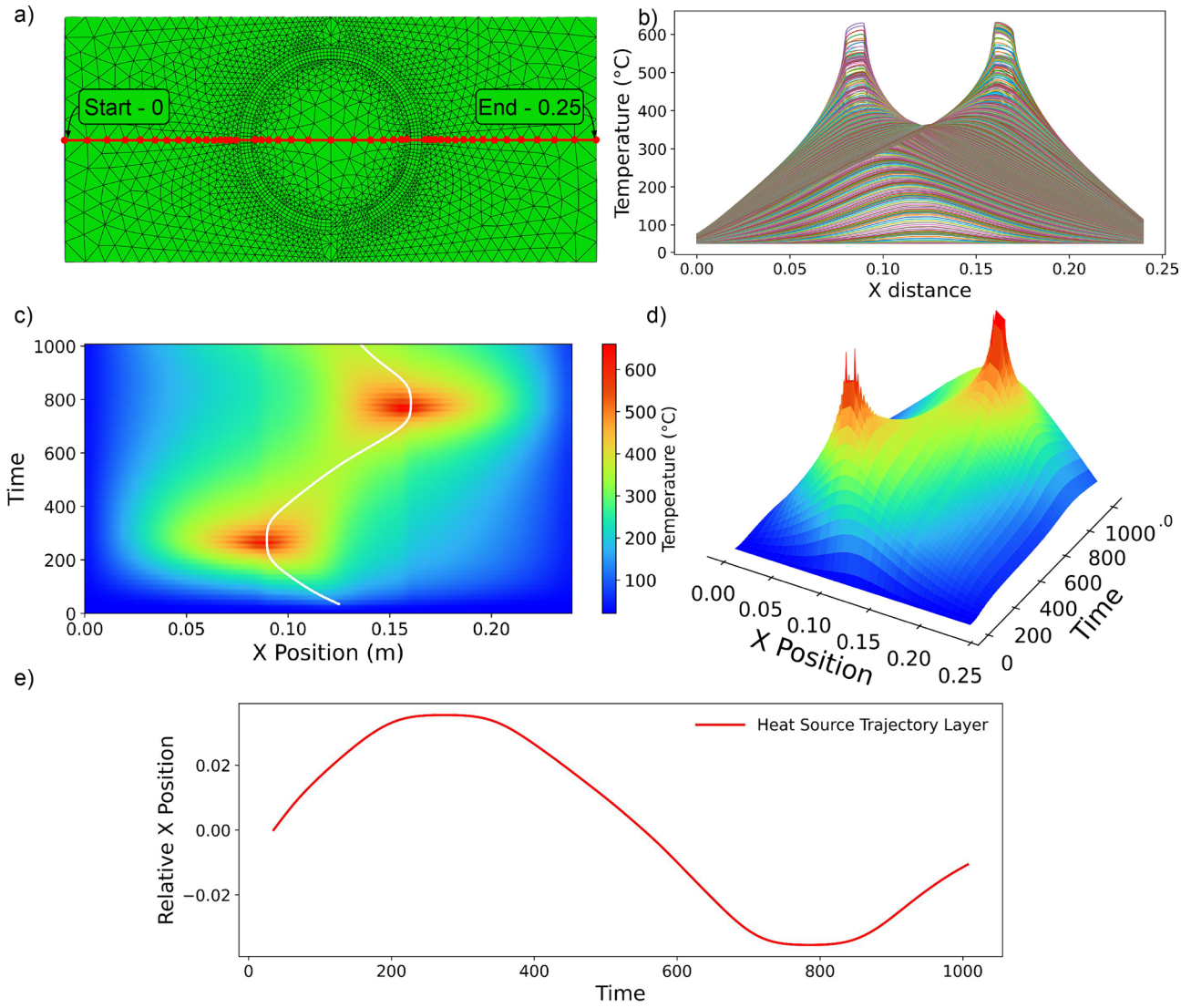


Fig. 11 Extraction and reconstruction of the transient temperature field for the second level hierarchical WAAM model. (**a**) Predefined extraction path along the substrate used for temperature data acquisition. (**b**) Temporal evolution of the thermal field along the selected path throughout all steps and frames. (**c**) Top view projection of the reconstructed spatio-temporal temperature surface with highlighted maximum temperature path. (**d**) Three-dimensional reconstruction of the transient temperature field, illustrating the propagation of the temperature peak and the associated temperature gradients during deposition. (**e**) Heat source trajectory obtained by tracking the position of the maximum temperature over time for layer level model.

dimensional temperature surface, Fig. 11d, and its corresponding top view, Fig. 11c providing a spacial temporal representation of heat propagation throughout the deposition cycle. Tracking the location of the maximum temperature on reconstructed surface yields the heat source trajectory, Fig. 11e. In contrast with firs level model, where it is represented by a constant straight line, the second level trajectory represents a periodic evolution governed by the circular deposition geometry. This demonstrates that the proposed reconstruction methodology captures the geometric characteristics of the deposition process independently of the underlying trajectory.

To facilitate comparison with subsequent hierarchical levels, the reconstructed trajectory was normalized with respect to its maximum lateral displacement, providing a dimensionless representation of the heat source motion. The normalized trajectory can be described by sinusoidal periodic function:

$$x(t) = sin(2\pi f t + \phi) \tag{24}$$

where $x(t)$ is the normalized heat source trajectory, t is time, f denotes the normalized oscillation frequency associated with one complete deposition cycle and ϕ is the initial phase. This analytical representation provides a compact mathematical description of the reconstructed hear source trajectory. The second hierarchical level extends trajectory description from linear bead level formulation to a periodic geometry dependent representation, establishing the basis for the mathematical formulation of the subsequent wall level model.

Using the temperature field obtained from the transient thermal analysis, the corresponding thermo-mechanical response of the deposited layer was evaluated, Fig. 12. Compared to the first level model, the interaction

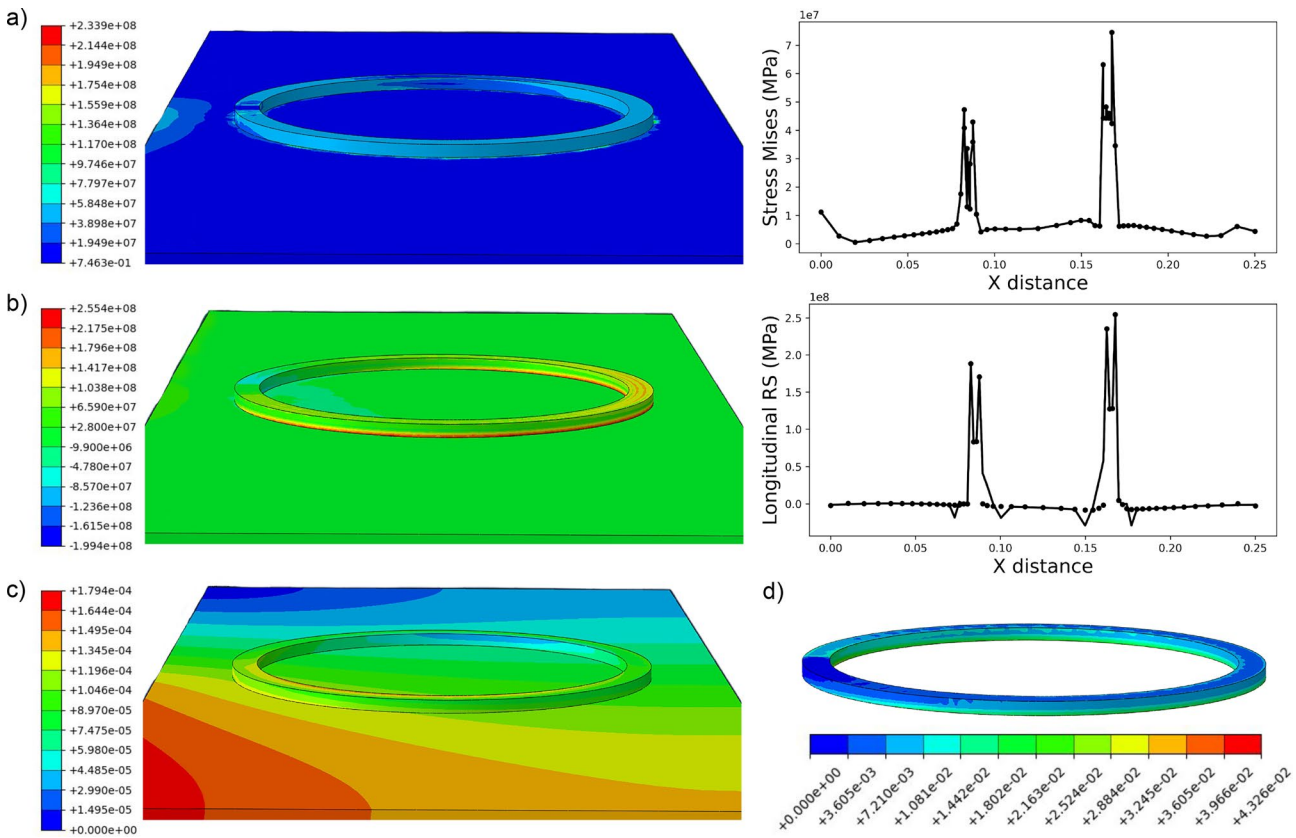


Fig. 12 Thermo-mechanical response of the second level hierarchical WAAM model. (**a**) Equivalent von Mises stress distribution together with the corresponding stress profile extracted along the predefined horizontal substrate path. (**b**) Longitudinal residual stress distribution with the corresponding path profile. (**c**) Displacement magnitude of the layer level model demonstrating distortion response. (**d**) Equivalent plastic strain distribution showing the accumulated plastic deformation within the deposited layer.

between sequentially deposited material generates a continuous stress field distributed along the circular deposition path due to heat accumulation during layer formation. The equivalent von Mises stress contour and the profile extracted along the predefined path Fig. 11a are showing localization of highest stress at bead-substrate interface Fig. 12a, with highest values of 234 MPa. Thermo-mechanical response remains concentrated beneath the deposited layer, stresses are decay rapidly with increasing distance from the heat source, resulting in 52-69 MPa. Longitudinal residual stress distribution is showing similar behaviour, Fig. 12b, with varying peak values of -199 MPa and 250 MPa. The repeated thermal expansion and contraction due to sequential deposition generate alternative tensile and compressive residual stress regions, reflecting the cumulative thermo-mechanical loading imposed. Analysis shows that loading is distributed relatively uniformly. Predicted distortion remains below 0.179 mm, Fig. 12c, provided that substrate gives mechanical restraint to suppress excessive deformation. The equivalent plastic strain reaches a maximum value of 4.3% in regions of bead-substrate interface, confirming that plastic deformation is localized within highest thermal gradients region. The results demonstrate a transition from highly centered thermo-mechanical response of a single bead to the collective mechanical behaviour of an entire deposited bead. The absence of extensive stress localization and widespread plastic deformation indicates the deposited layer preserves its structural integrity under adopted processing conditions, providing a mechanically stable foundation for the third level simulation.

Third level numerical results

The third hierarchical level resolves the transient thermal evolution of the wall or complete desired manufacturing component throughout sequential material deposition Fig. 13. Beads are forming layers and wall subsequently. Three representative steps were chosen to demonstrate temperature distribution obtained through the

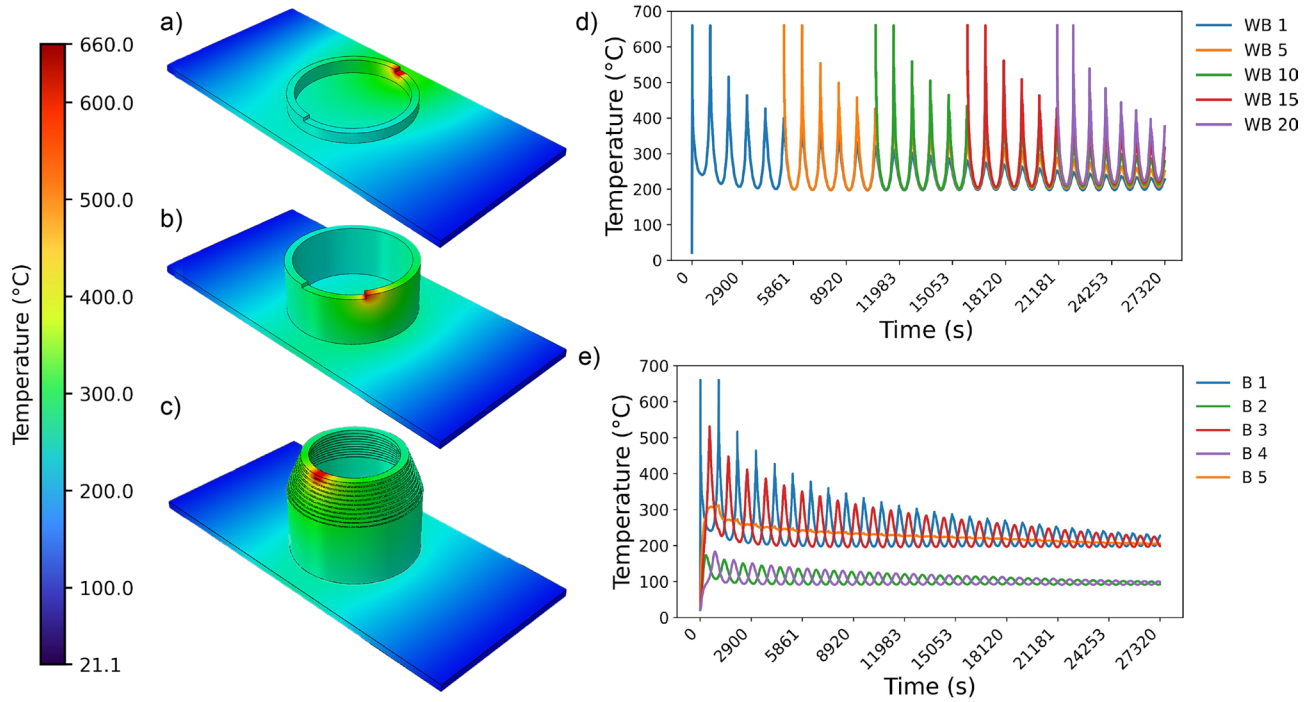


Fig. 13 Thermal response of the third level hierarchical WAAM model. Predicted temperature distribution of wall deposition process at Step 262 (**a**), Step 1381 (**b**) and Step 2733 (**c**) showing progressive evolution of the thermal field during sequential layer deposition. (**d**) Temperature histories recorded at the selected wall welding beads monitoring points WB1, WB5, WB10, WB15, WB20 representative of different global heights illustrating repeated thermal cycling and progressive heat accumulation. (**e**) Temperature histories recorded at the substrate monitoring points B1-B5, demonstrating gradual evolution of heat accumulation and thermal dissipation.

process. Step 262 which represents initial stage of the process Fig. 13a, Step 1381 - intermediate stage Fig. 13b and at Step 2733 - final stage Fig. 13c. Data demonstrates progressive thermal field expansion into the component with increasing hight of manufacturing component. In contrast to the bead and layer levels, where phenomena was mainly localized to the intersection regions of bead and substrate, at the third level most of material do not dissipate heat directly into the substrate but using previously deposited material as a transfer material promoting retention of residual thermal energy between deposition cycles. Applied boundary heat sink of 21.1°C continuously removing conducted thermal energy from the computational domain not letting the deposited material reach critical values of the melting point. The resulting temperature field therefore represents a transient balance between volumetric heat input associated with the moving heat source and boundary controlled heat dissipation. The substrate monitoring points B1-B5, Fig. 6e, data presented at Fig. 13d are showing similar results to the second level behaviour. Points that are more distant from moving heat source B2 and B4 experiencing lower heat values, central point B5 is also showing gradual heat accumulation along the deposition time. Melting temperature reached by points that are closest to the deposition location B1 and B3 are in fact, nothing more but the limitation of the mesh produced Fig. 6d, represented by the element that is coincide with the bead deposition region and is not representing correctly real life peak temperature values. For the the third level simulation, all the information from 27 welding beads data points Fig. 6f was extracted. Amount of data records for one these particular points exceed $N_{WB_n} = 49678$ cells, total 27 points data results in $N_{WB_{total}} = 1341306$ records. Displaying such an extensive amount of data is unnecessary. Therefore five representative points WB1, WB5, WB10, WB15 and WB20 were chosen Fig. 13e to demonstrate transient heat evolution at different wall heights. Each monitoring location exhibit rapid heating during direct interaction with the moving heat source followed by conductive cooling into the surround material, that might be the substrate, neighboring beads or the beneath deposited layers, generating successive reheating cycles and elevated thermal baseline. The gradual reduction of temperature oscillation amplitude observed along the time indicates that global thermal field progressively approaches a quasi-periodic transient regime, where rate of thermal energy introduced by the moving heat source becomes increasingly balanced by conductive heat dissipation. Adopted deposition strategy prevents uncontrolled thermal accumulation and maintains a thermally stable manufacturing process.

To further investigate the evolution of heat propagation during wall fabrication, the transient temperature field was reconstructed from the predefined substrate extraction path Fig. 14a. The temperature profiles obtained for all analysis steps and frames, Fig. 14b were assembled into a three dimensional spacial temporal representation of the thermal field, Fig 14c, while the corresponding top view Fig. 14d enabled reconstruction of the heat source trajectory by tracking the local temperature peaks. Unlike first and second level models, this level's model exhibit a progressive reduction of maximum temperature during oscillatory movement determined by the circular geometry of the desired component. To quantify this behaviour, maximum temperature T_i of the time interval i associated with each oscillation O_i and global maximum temperature T_{max} were extracted and used to normalize the oscillation amplitude according to:

$$A_i = A_0 \frac{T_i}{T_{max}} \tag{25}$$

where A_0 is initial oscillation amplitude and A_i is the current one. The normalized heat source trajectory for each oscillation can be defined as:

$$x(t) = A_i \sin(2\pi f t + \phi), \quad t \in \mathcal{O}_i \tag{26}$$

And complete normalized heat source trajectory for the third level wall with progressively varying amplitudes:

$$x(t) = \begin{cases} A_1 \sin(2\pi f t + \phi), & t \in \mathcal{O}_1 \\ A_2 \sin(2\pi f t + \phi), & t \in \mathcal{O}_2 \\ \vdots \\ A_n \sin(2\pi f t + \phi), & t \in \mathcal{O}_n. \end{cases} \tag{27}$$

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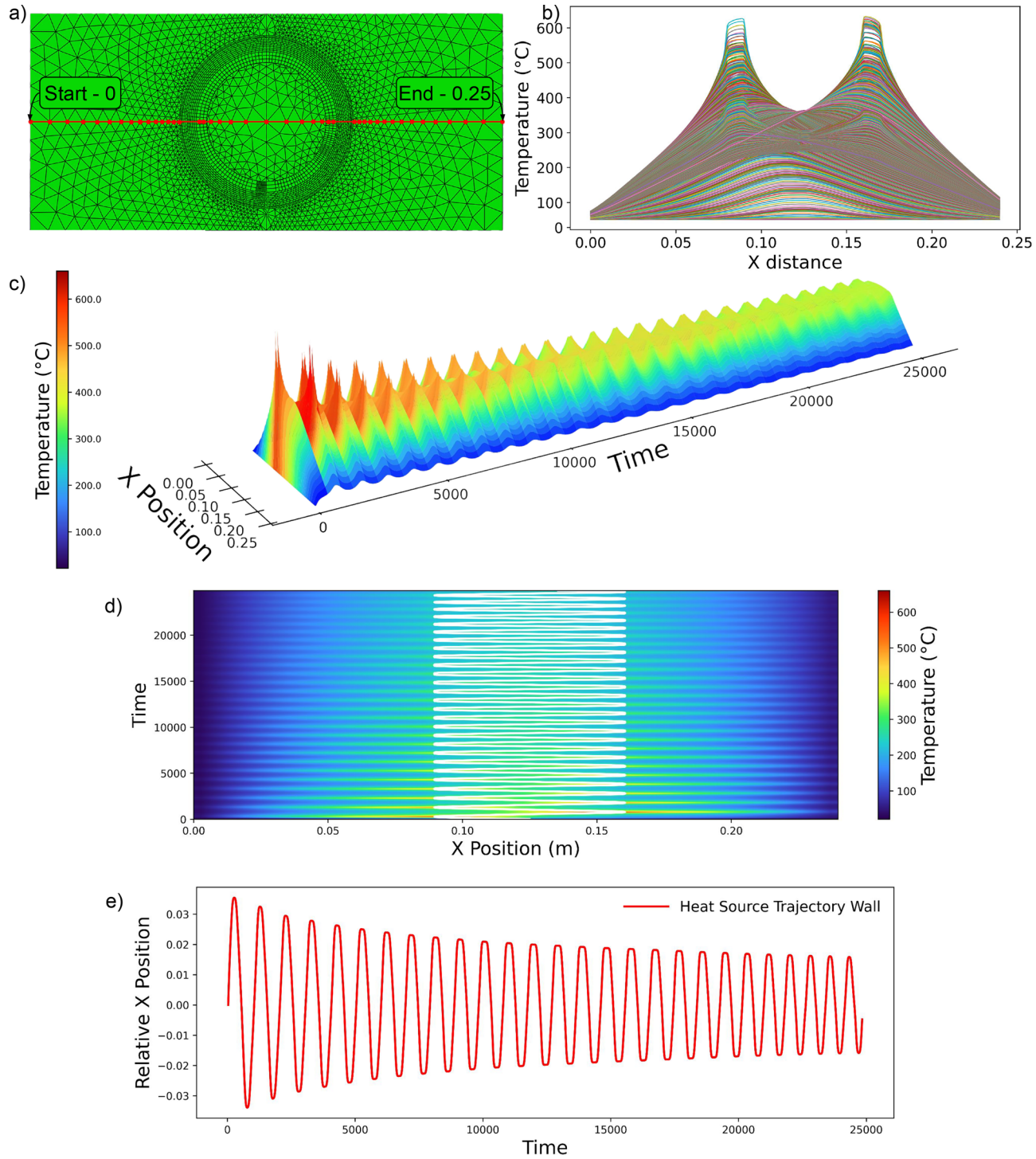


Fig. 14 Extraction and reconstruction of the transient temperature field for the third level hierarchical WAAM model. (**a**) Predefined extraction path along the substrate used for temperature data acquisition. (**b**) Temporal evolution of the thermal field along the selected path throughout all steps and frames. (**c**) Three-dimensional reconstruction of the transient temperature field, demonstrating the heat accumulation and peak reduction throughout 2734 steps. (**d**) Top view projection of the reconstructed spatio-temporal temperature surface with highlighted maximum temperature path. (**e**) Heat source trajectory obtained by tracking the position of the maximum temperature over time for wall level model, with oscillation amplitude normalized by ration of T_i/T_{max}.

This formulation directly links the reconstructed trajectory to the transient thermal response, allowing the oscillation amplitude to evolve accordingly to the temperature history nd the changing deposition geometry. Furthermore, the evolution of the normalized heat source trajectory can be approximated by the continuous function, Fig. 15, Eq. 28:

$$x(t) = A_0 e^{-kt} \sin(2\pi f t + \phi) \tag{28}$$

where k is exponential decay and ϕ is the phase shift. The exponential formulation provides a compact analytical approximation description of the evolution of the normalized heat-source trajectory.

The generalized heat-source trajectory can also be used as a reference for process planning and future adaptive control. Its parameters provide a compact description of the expected heat-source motion and thermal evolution, allowing the planned trajectory to be compared with real-time temperature measurements and, consequently, supporting future adjustment of process parameters such as heat input, travel speed, or cooling conditions. The formulation may further serve as a reduced representation of the thermal state for integration with digital-twin systems. These applications are considered future extensions of the present framework.

The thermo-mechanical response of the third level hierarchical WAAM model Fig. 16 demonstrating, that stress is no longer confined to bead and substrate interaction but is also redistributed along the whole component. The equivalent von Mises stress Fig. 16a is found to be more pronounced , approximately 76 MPa, along the height of the component. Concentrating mostly within upper deposited layers and near substrate edges situated near deposited wall as a consequence of thermal loading. The corresponding profile along extracted path, previously defined Fig. 14a, demonstrating a typical response with average broad maximum reaching around 102 MPa at highest thermally influenced region followed by gradual decrease towards to the heat dissipation lateral regions. The longitudinal stress distribution demonstrates mostly compressive response Fig. 16b, varying around 37MPa, reflecting constrained thermal contraction during successive deposition and cooling cycles. Compared to localized residual stresses at lower hierarchical levels, wall level predicts a a continuous residual stress field extending throughout the component. Although global maxima of tensile stress reaches approximately 168 MPa and is found at the wall substrate interface, it is expected, and overall global model is smoothly uniform not reaching high stress values. Extracted profile showing analogical to von Mises stress with stress distributed mostly at the region of higher temperature influence. Predicted displacement Fig. 16c is found to be limited to the higher regions and develops as a global distortion of the manufactured component. The highest values of equivalent plastic strain Fig. 16d remain confined to regions of the highest thermo-mechanical loading in substrate wall interface while the remaining volume undergoes predominantly elastic deformation. The predicted thermo-mechanical response indicates a stable redistribution of stresses and strains throughout the completed component

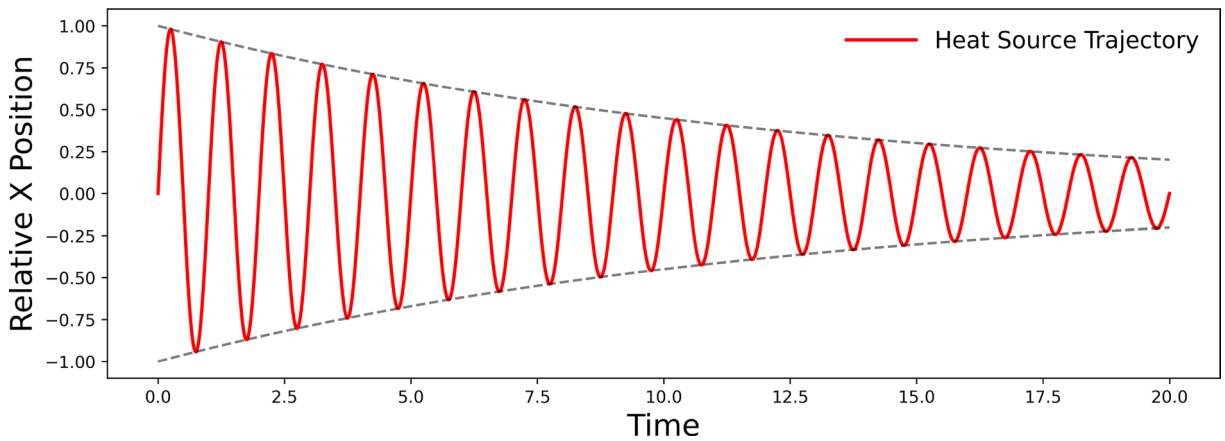


Fig. 15 Continuous exponential approximation of the normalized heat-source trajectory with progressive reduction of oscillation amplitude for the third level hierarchical WAAM model.

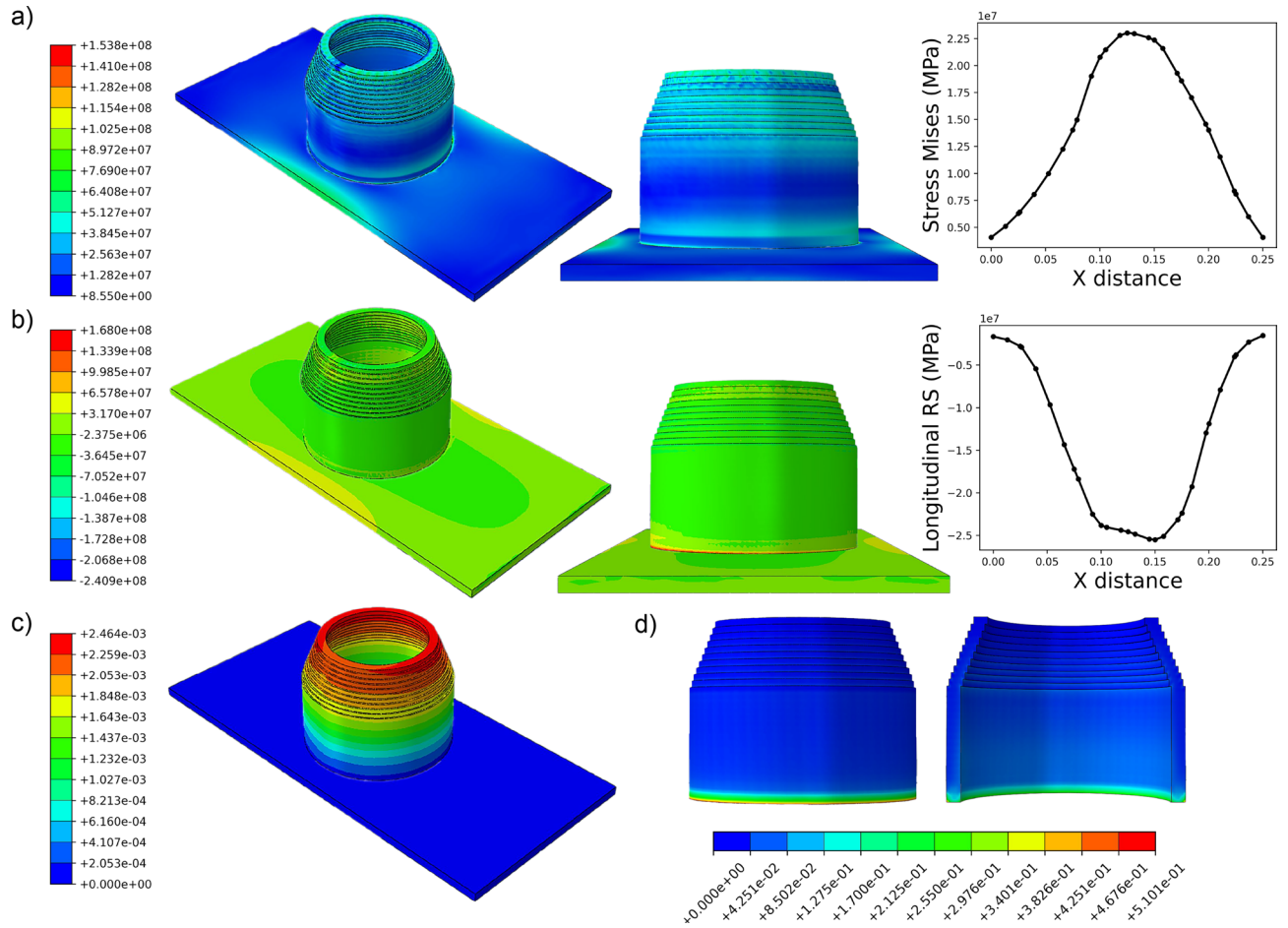


Fig. 16 Thermo-mechanical response of the third level hierarchical WAAM model. (**a**) Equivalent von Mises stress distribution, isometric and front views, together with the corresponding stress profile extracted along the predefined horizontal substrate path. (**b**) Longitudinal residual stress distribution, isometric and front views, with the corresponding path profile. (**c**) Displacement magnitude of the completed component, isometric view, demonstrating global deformation response from thermo-mechanical loading. (**d**) Equivalent plastic strain distribution, front and sectional views, showing the localization of accumulated plastic deformation within the deposited wall.

without evidence of extensive plastic localization of excessive stress concentrations that would compromise the simulated manufacturing process.

Computational cost

The computational cost associated with each hierarchical level was recorded. All numerical simulations were performed on a workstation equipped with an Intel Xeon processor, 32GB RAM and an NVIDIA GeForce GTX 960. Shared memory parallelization of 16 CPU threads was employed for element level computations, global matrix assembly and the solution of sparse linear systems. CUDA based GPU acceleration of 1024 cores provided was used for sparse matrix factorization and back-substitution. The complete thermo-mechanical analysis generated more than 1.6 TB of output data, requiring continuous storage of transient field variables results. The computational cost increased progressively with the complexity of the hierarchical model, Table 4. Although bead and layer level analyses were completed within relatively short time, the wall level simulation dominated the overall computational effort due to the large number of transient increments, activated elements and non-linear thermo-mechanical equilibrium iterations. Throughout all levels the structural analysis consistently required

Table 4 Computational performance of the three-level hierarchical WAAM model on Intel Xeon workstation with 32 GB RAM, 16 CPU threads and an NVIDIA GeForce 960 GPU.

Hierarchical level	Thermal analysis	Structural analysis	Total
Bead	43 s	2 min 7 s	2 min 50 s
Layer	23 min 31 s	1 h 6 min 53 s	1 h 30 min 24 s
Wall	12 h 11 min 59 s	4 d 8 h 25 min 58 s	4 d 20 h 37 min 57 s
Overall	–	–	4 d 21 h 58 min 11 s

longer solution times than the thermal analysis. Despite the high computational demand associated with the full wall simulation, the present three-level hierarchical framework substantially improves computational efficiency during process development. The first and the second levels are enabling rapid validation and optimization of fundamental deposition parameters before executing the computationally expensive third level analysis, helping to avoid repeated full scale expensive simulations. Compared with conventional single scale thermo mechanical modelling strategies reported in literature[57], the present framework reduces both computational time and storage requirements while maintaining comparable modelling fidelity[58]. Developed model provides a practical balance between computational efficiency and simulation fidelity for large scale WAAM components.

The bead and layer levels required only 1 h 33 min 14 s, approximately 1.34% of the wall-level computational time. Thus, these lower levels provide an efficient means of screening process parameters before performing the computationally intensive full-wall simulation. For three candidate configurations, this strategy can reduce the estimated computational effort by approximately 66% compared with repeated full-scale simulations.

Experimental WAAM results

Numerically simulated desired aluminium component was experimentally validated using the three-level hierarchical WAAM model. The experimental arrangement of the robotic WAAM system described earlier in the Methodology section, is presented in Fig. 17b.

Quantitative comparison between the numerical prediction and experimental measurements is summarized in Table 5. The experimental validation was limited to the component geometry and transient temperature response, as no experimental measurements of residual stress or distortion were available. The manufactured component Fig. 17a has been reproduced with high accuracy following the target geometry. The measured bottom diameter is 84.375 mm and measured height is 67.313 mm, corresponding to relative deviations of only 0.74% and 0.28 % . Manufacturing process took total of 27 layers, varying in height between 2.3 mm and 2.7 mm and width of 4.8 mm and 5.3 mm remaining in close agreement with predicted values. Experimentally obtained temperature data was compared with numerically predicted,

Figure 17c, at the same chosen monitoring point. Experimental measurements have reproduced the characteristic cyclic thermal behaviour predicted. The peak temperature has reached 667°C compared with the predicted 660° C deviating only about 1.05 %. The slightly higher peak temperature observed experimentally is consistent with the physical melting behaviour of aluminium, where complete fusion requires temperatures marginally exceeding the nominal melting temperature. Initially, observed peaks and lows of the temperature were slightly higher than the predicted ones, however, as deposition progressed, the experimental thermal history converged towards the numerical prediction, indicating that the model accurately captured the transient heat accumulation and subsequent thermal stabilization of the process. The close agreement between the numerical predictions and experimental measurements demonstrates that the present hierarchical WAAM framework accurately reproduces both the transient thermal response and the resulting geometric characteristics of the manufactured component. These results are confirming the model suitability for process parameter prediction and experimental implementation.

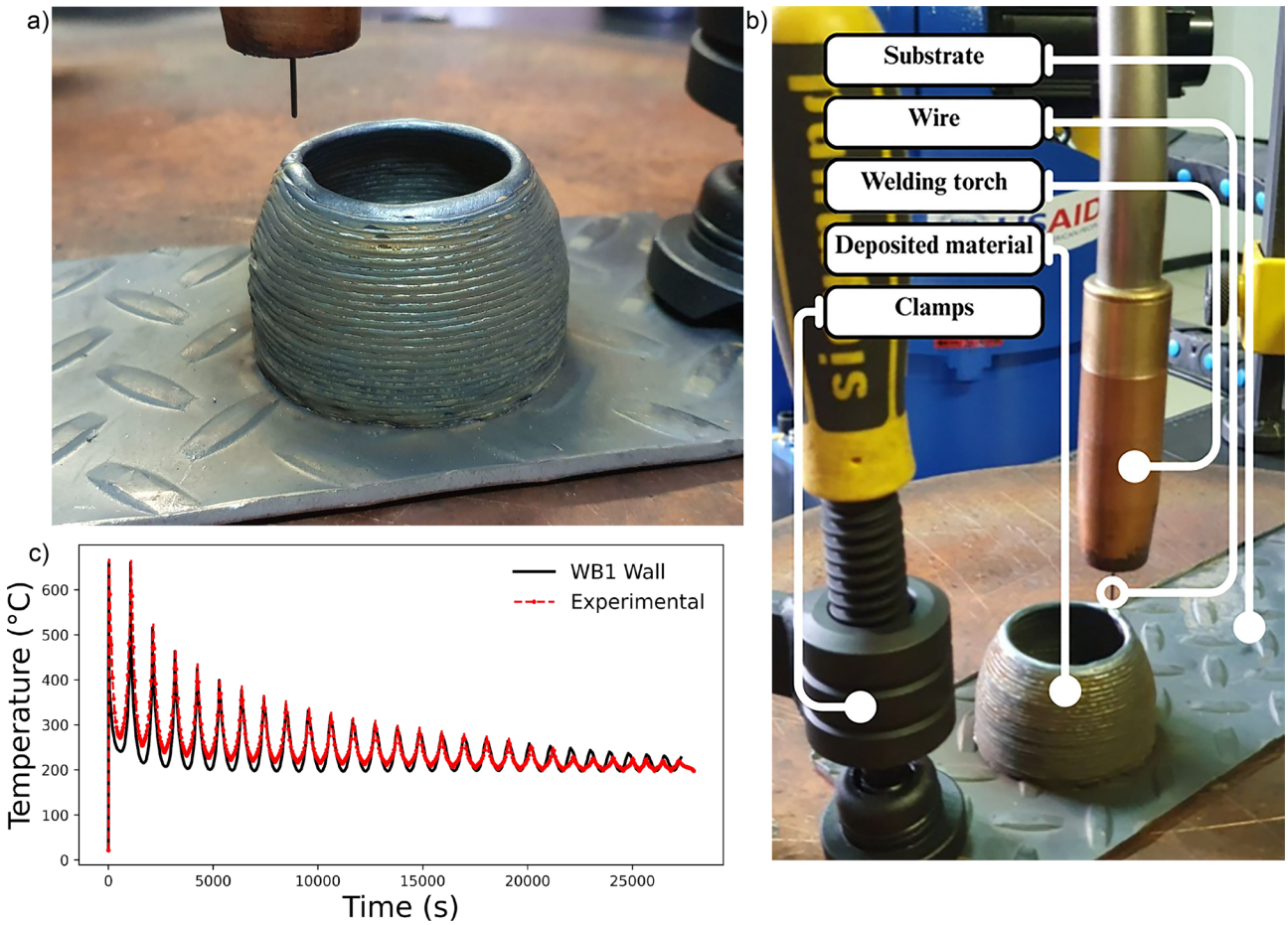


Fig. 17 Experimental validation of the desired component using proposed three-level hierarchical WAAM model. (**a**) WAAM manufactured component. (**b**) WAAM system experimental arrangement setup showing already fabricated component. (**c**) Comparison between numerically predicted and experimentally measured temperature histories at monitoring point WB1.

Table 5 Comparison between the numerical predictions and experimental measurements of the fabricated WAAM component.

Parameter	Numerical model	Experimental result	Relative error
Bottom outer diameter	85.000 mm	84.375 mm	0.74%
Obtained height	67.500 mm	67.313 mm	0.28%
Number of layers	27	27	0.00%
Layer height	2.5 mm	2.3–2.7 mm	±8.0%
Bead width	5.0 mm	4.8–5.3 mm	−4.0 to +6.0%
Peak temperature	660°C	667°C	1.05%

Conclusion

A three-level hierarchical WAAM modelling framework has been used for numerical prediction and experimental validation of an aluminium nozzle-like circular component, together with a mathematical formulation of the heat-source trajectory. Unlike conventional single-scale approaches, the present framework decomposes the manufacturing process into bead, layer and wall levels, allowing each hierarchical stage to resolve the dominant thermo-mechanical phenomena. This enables the progressive evaluation of thermo-mechanical behaviour, thermal accumulation, residual stress and geometric evolution. A mathematical heat-source trajectory formulation

was also developed to represent the deposition paths at the different hierarchical levels. The wall-level model reproduced the complete manufacturing sequence of 27 layers and 1,431 sequentially activated beads.

Experimental validation showed good agreement between the numerical predictions and the manufactured component. The predicted bottom outer diameter of 85.000 mm and height of 67.500 mm differed from the experimental values of 84.375 mm and 67.313 mm by 0.74% and 0.28%, respectively. The predicted peak temperature of 660°C compared with the experimentally measured 667°C, corresponding to an error of 1.05%. The predicted 27-layer deposition sequence was also consistent with the experiment, while the measured layer height of 2.3–2.7 mm and bead width of 4.8–5.3 mm were close to the predicted values of 2.5 mm and 5.0 mm. These results demonstrate the capability of the proposed framework to reproduce the principal thermal and geometric characteristics of the investigated component.

Several limitations should nevertheless be acknowledged. The simplified uniform heat-source model does not fully represent detailed melt-pool physics, while the validation was limited to aluminium 6061 and a single component geometry. Experimental validation was primarily based on geometry and thermal history, and the wall-level simulation remains computationally demanding. Future work will therefore consider alternative heat-source models, computational fluid dynamics, micro-structural simulations, real-time sensor feedback and adaptive process control, with the longer-term objective of developing more autonomous and reliable WAAM systems.

Author contributions P.A. conceptualized the study and developed the methodology. P.A. and F.B. performed the numerical modeling and simulations. P.A. and G.S. conducted the experimental work and data acquisition. P.A., F.B. and G.S. carried out the validation and formal analysis. P.A. wrote the original draft of the manuscript. All authors reviewed, edited, and approved the final manuscript.

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Declarations

Competing interests The author(s) declare no competing interests.

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